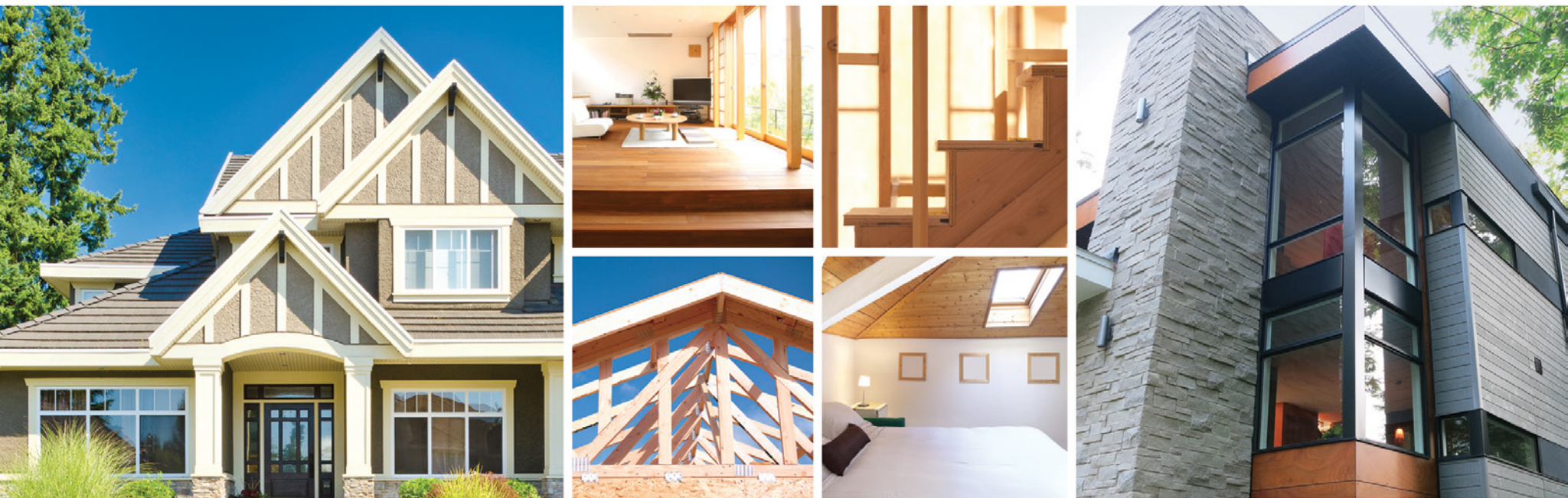


CANADIAN WOOD-FRAME HOUSE CONSTRUCTION



Canada 


CMHC  SCHL
HOME TO CANADIANS

CHAPTER 11

Ceiling and Roof Framing



A roof must have a water-shedding surface air and vapour barriers to prevent moisture accumulation in the roof assembly, thermal insulation and ventilation to remove moisture that has escaped from the conditioned space. In Canada, most houses are framed with prefabricated wood trusses and have an unheated, ventilated attic space. Cathedral ceilings and flat roofs do not have attic spaces, but there must still be a vented space above the insulation to remove moisture.

Steeper roofs tend to shed water better than flat or low-slope roofs. The slope of a roof is expressed as a rise-to-run ratio with the vertical component, or rise, always shown first. There are two conventions for expressing the slope of a roof: imperial and ratio.

The imperial convention is based on the use of a framing square, and the run is always expressed as 12, based on 12 inches to a foot. For example, a roof with a slope

of 45° is expressed as a $1\frac{1}{2}$ slope. A roof with a $\frac{4}{12}$ slope has a rise of 4 inches for every 12 inches of run.

When expressed as a ratio, for slopes less than 45° , the *first* number (the vertical component) is always shown as one. A ratio of 1:5, for example, indicates a vertical of 1 unit for every 5 units of horizontal dimension. For slopes steeper than 45° , the *second* number (the horizontal component) is always shown as one. A ratio of 5:1, for example, indicates a vertical of 5 units for every one unit of horizontal dimension. The use of mixed units such as 1 mm in 10 m should be avoided.

Expressed as a ratio, the imperial designation of $\frac{4}{12}$ is 1:3. In special cases where a high degree of accuracy is required, angular expressions of slope are acceptable.

Pitched roofs vary in slope from 1:6 to 1:1 or more. Roof slope affects such factors as the minimum slope required for various

roofing materials to reduce the risk of water backing up through the roof membrane. The minimum allowable slope for asphalt shingles is 1:6, and there are different installation requirements for roof slopes of 1:3 or greater and slopes less than 1:3. There are minimum allowable slopes for each type of roofing material.

The dimensions of roof joists and rafters for the various grades and species of lumber and for various live loads are given in Tables 29 to 32 (p. 295-301).

PITCHED ROOFS

Prefabricated roof trusses are most often used for framing residential roofs. Rafters framed on site are sometimes used. Of the pitched roofs, the gable roof is the simplest to construct. Other pitched configurations, such as a hip or L-shape roof, can also be framed with trusses (Figure 77).

Prefabricated Roof Trusses

Prefabricated roof trusses offer many advantages. They are designed to handle snow loads that the trusses will be subjected to, are manufactured

77

L-shape trussed roof

See isometric below

ridge
valley trusses
roof sheathing
gable end
girder truss
common trusses
double wall plates

Note: For clarity, some structural members of some of the trusses have been omitted, and roof sheathing appears continuous.

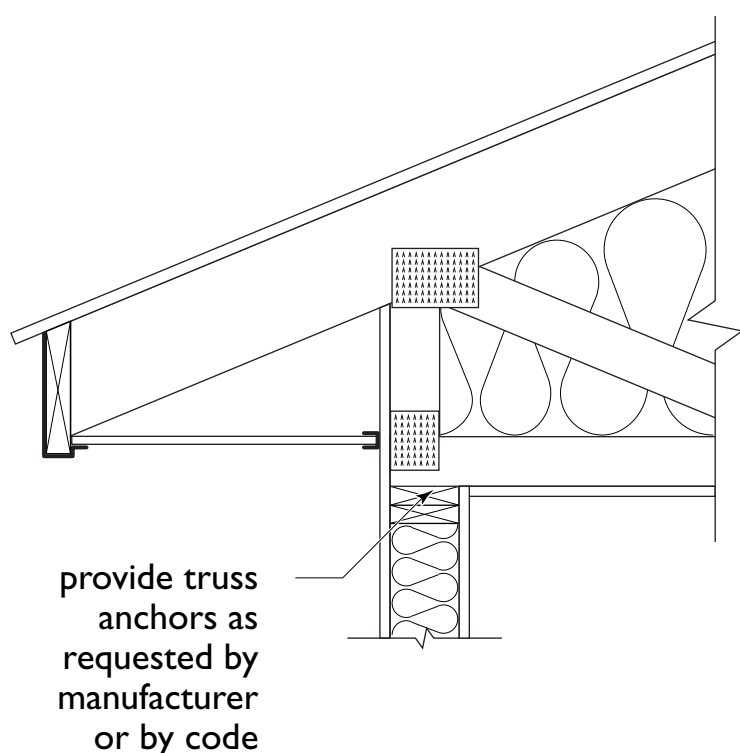
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Ceiling and Roof Framing

with a high level of quality control, reduce lumber waste and speed up the process of enclosing the house. They provide a surface for the roof sheathing, a surface for the ceiling finish material and a space for insulation. Raised heel trusses (Figure 78) are used in most new houses because they provide a deeper space at the eaves to accommodate greater amounts of insulation. The energy efficiency requirements in most building codes lead to the use of trusses, rafters and roof joists with a vertical dimension at the outer face of the exterior wall framing of 300 mm (12 in.) or more.

Ventilation of the attic space is easily provided through the eaves and at the ridge. In most cases, trusses are designed to span from exterior wall to exterior wall with no intermediate load-bearing walls to support the roof loads (Figure 79). This gives flexibility to interior planning because partitions can be placed without regard to structural requirements. A continuous air barrier and vapour retarder can be applied to the underside of the trusses before the interior partitions are erected.

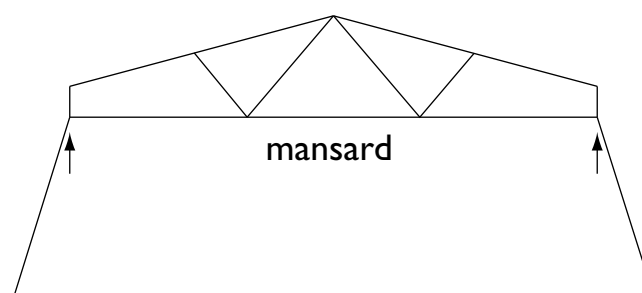
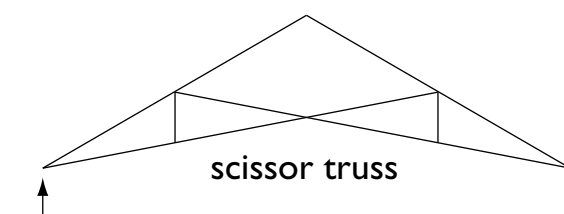
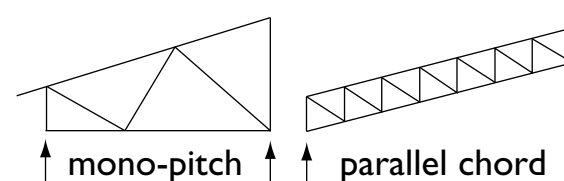
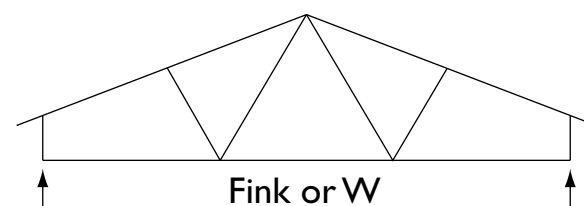
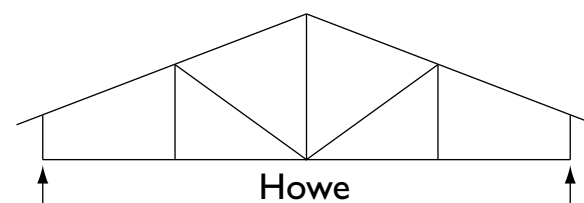
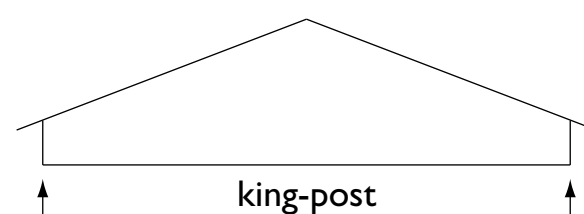
78 Raised heel roof truss provides depth for insulation



Prefabricated trusses should be stored on a flat, clean area of the construction site. Trusses shorter than a 6 m (20 ft.) span are usually installed by hand. Trusses longer than this require special lifting techniques to avoid damage.

Trusses must be lifted into a vertical position carefully to prevent excessive lateral bending. The gable truss is placed first and braced to

79 Types of prefabricated roof trusses

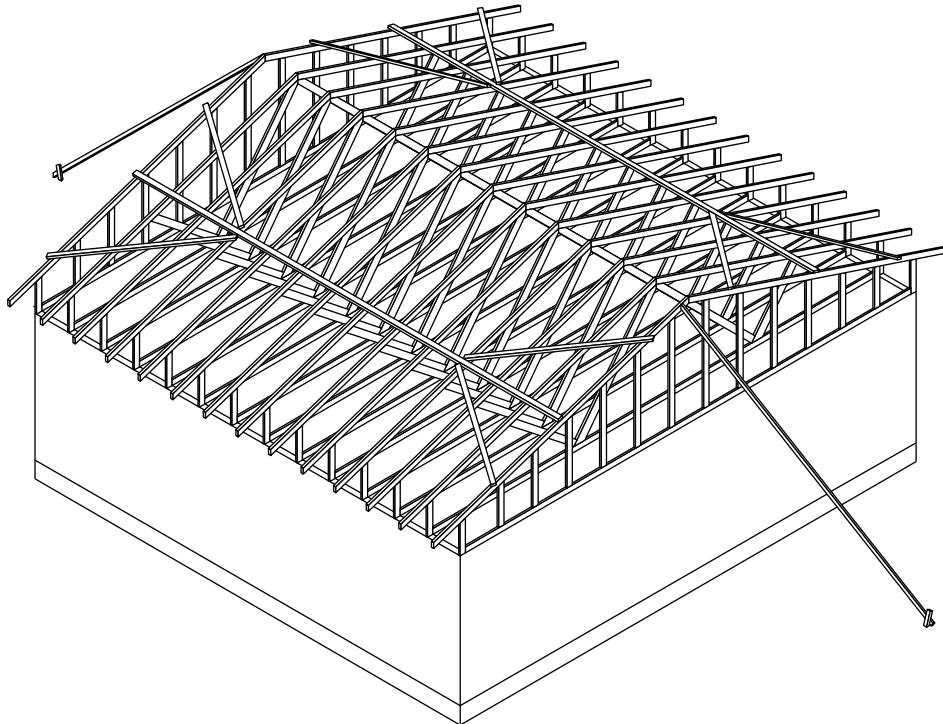


CHAPTER 11
Ceiling and Roof Framing

the ground and wall. Each additional truss is lifted into position, generally 600 mm (24 in.) on centre, toe-nailed to the top plates and

temporarily braced (Figure 80). When all trusses are plumbed and properly positioned, they are braced permanently (Figure 81); a) permanent

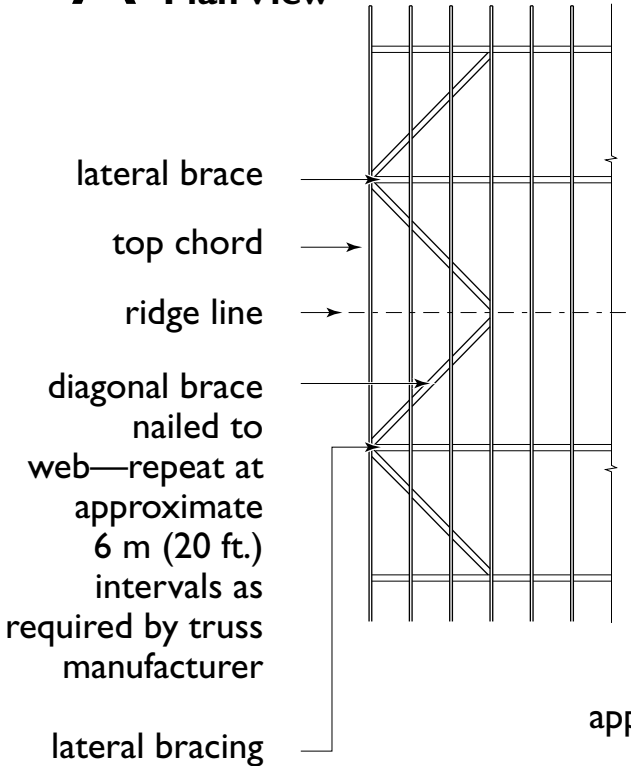
80 Temporary bracing of roof trusses



Note: Gable ends are normally sheathed prior to installation of end braces.

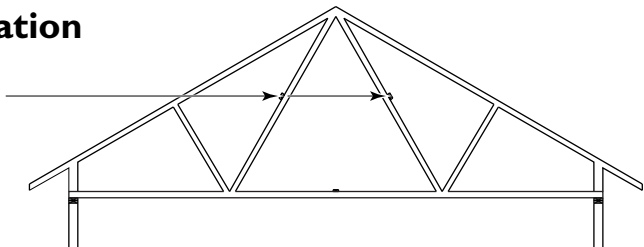
81 Permanent bracing of roof trusses

A Plan View



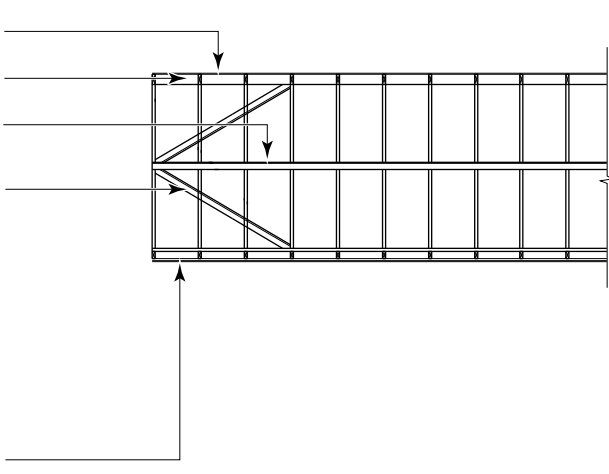
B End Elevation

38 x 89 mm
(2 x 4 in.) web
bracing as
required by truss
manufacturer



C Side Elevation

sheathing
blocking
lateral brace
diagonal forms
braced bay—repeat
at both ends and at
approximate 6 m (20 ft.)
intervals as required
by truss manufacturer
ceiling



bracing of top chord plane, b) lateral web bracing and c) permanent bracing to web member or bottom chord plane. The stiffness of the roof is increased when the sheathing has been applied. Trusses must be installed and braced according to the manufacturer's instructions and must not be cut or altered.

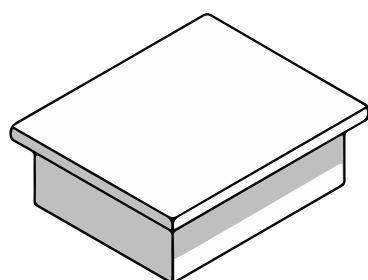
Site Assembly of Pitched Roofs

For on-site construction, the gable roof is the simplest roof constructed with rafters or trusses (Figure 82), which are the same length and pattern and can be erected rather easily. Dormers may be added to provide additional light, headroom and ventilation (see Figure 88). The hip roof design uses common rafters fastened to a ridge board, with hip rafters providing the support for the jack rafters (Figure 83b).

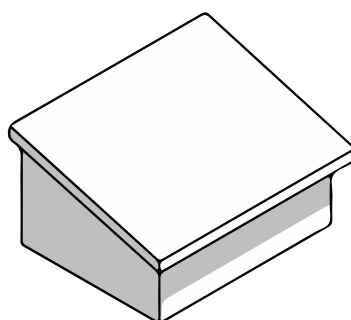
Liveable attic spaces need adequate insulation, an air barrier system and a vapour barrier as discussed in Chapter 5. The size of framing members found in Tables 29 to 32 (pp. 295-301) may need to be increased to provide space for additional insulation to meet energy efficiency requirements, or additional wood framing may be added to increase the space in the roof cavity.

Ceiling joists are used to support the ceiling finish and to act as ties between exterior walls and, in some cases, opposing rafters. They may also provide support for roof loads transferred to them by dwarf walls (knee walls) used as intermediate support for rafters, in which case they must be sized accordingly. (See Table 33 on p. 303 for ceiling joist spans.) When ceiling joists also support floor loads, determine what size is needed from the floor joist tables. (See Table 19 and Table 20 on pp. 283-285)

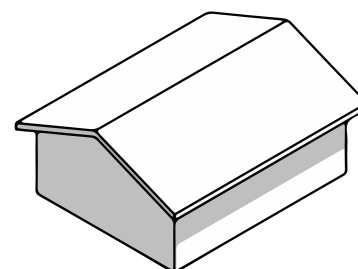
82 Types of pitched roofs



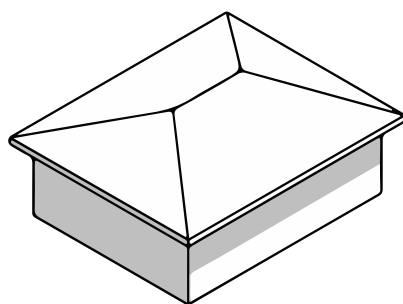
Flat



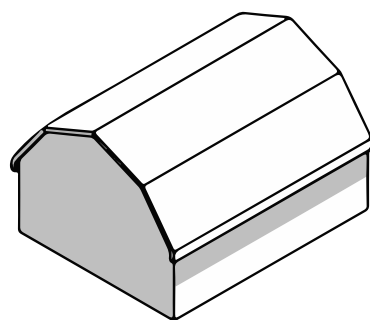
Monopitch



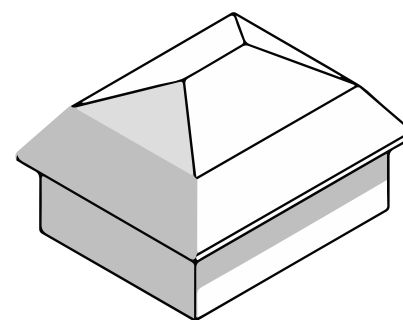
Gable



Hip



Gambrel



Mansard

CHAPTER II
Ceiling and Roof Framing

For pitched roofs framed with dimension lumber, nail the ceiling joists in place after the interior and exterior wall framing is complete but before the rafters are erected, because otherwise the thrust of the rafters will tend to push out the exterior walls. Ceiling joists are generally used to tie the lower ends of the rafters in pitched roofs that slope 1:3 or more. To prevent the rafter ends from moving outward, nail the ceiling joist to the side of each pair of rafters (Figure 83). For hip roofs

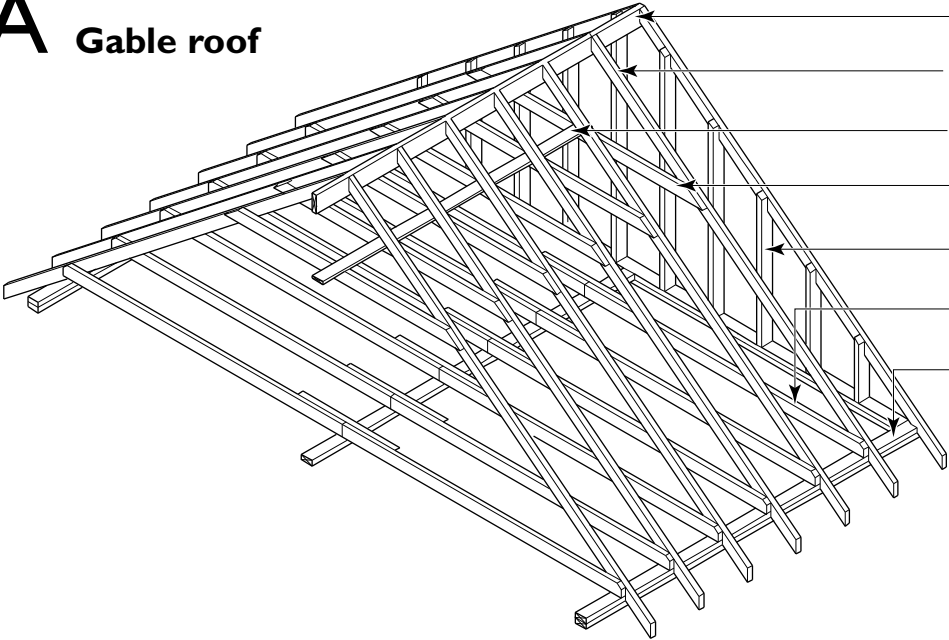
where jack rafters run perpendicular to the ceiling joists, a stub joist can tie the jack rafter back to the ceiling joist (Figure 85). Lap and nail the joists together or splice them at the centre load-bearing wall to provide a continuous tie across opposing rafters. The number of nails required in the connections depends upon the roof slope, rafter spacing, snow load and the width of the house. (See Table 24 on p. 288 for nailing requirements.)

83

Roof framing and attachment

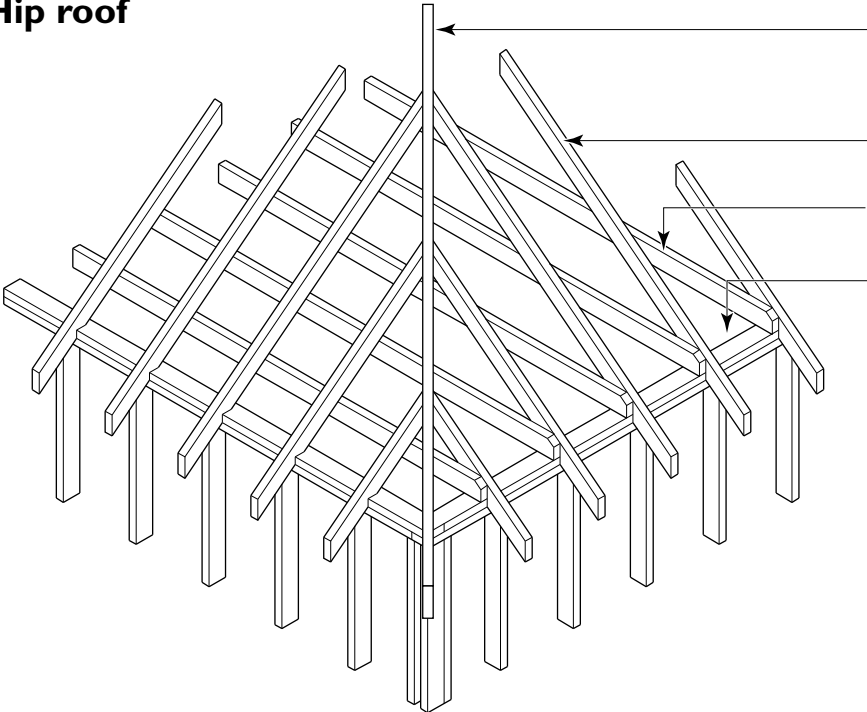
A

Gable roof



B

Hip roof



CHAPTER 11
Ceiling and Roof Framing

The additional roof load imposed by dwarf walls that run at right angles to ceiling joists (Figure 84) must be considered when choosing the size of the joists. An increase in the depth of the joists to the next standard depth will usually provide the additional strength required where the roof slope is more than 1:4. When the slope is 1:4 or less, determine the size of the ceiling joists from the span tables for roof joists (Tables 29 and 30 on pp. 295-297).

Since hip rafters are usually about 51 mm (2 in.) deeper than common or jack rafters, they reduce the space along the end walls to the extent that in low-slope roofs, there may not be enough room to install the outside ceiling joist at normal spacing from the wall. In this case, doubled joists are used and positioned to suit the available space (Figure 85). Tail joists are then added and toenailed to the outside wall plate and end-nailed

84

Rafter heel supported on a rafter plate

Rafter plate nailed to top of each ceiling joist with at least two 100 mm (4 in.) nails

rafter plate

rafter

ridge beam

exterior wall

knee wall with top and bottom plates and stud in line with each ceiling joist and rafter

struts at 1.2 m (4 ft.) o.c.

ceiling joists lapped directly above interior load-bearing wall

85

Doubled ceiling joists and stub joists used where a hip rafter reduces clear space near the end wall

jack rafter

hip rafter at least 50 mm (2 in.) deeper than jack rafters

ceiling joist

double ceiling joist

normal location for ceiling joist if space is available

stub joist

end wall framing top plate

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Ceiling and Roof Framing

to the doubled joist. The spacing of these tail joists is usually the same as that of the main ceiling joists.

Cut rafters to length with the proper angle at the ridge and notches (known as bird's mouth joints) on the eaves. The heel or lower part of the rafters should bear directly over the exterior wall. Depending on the design of the roof and the shape of the outside walls, place the rafters:

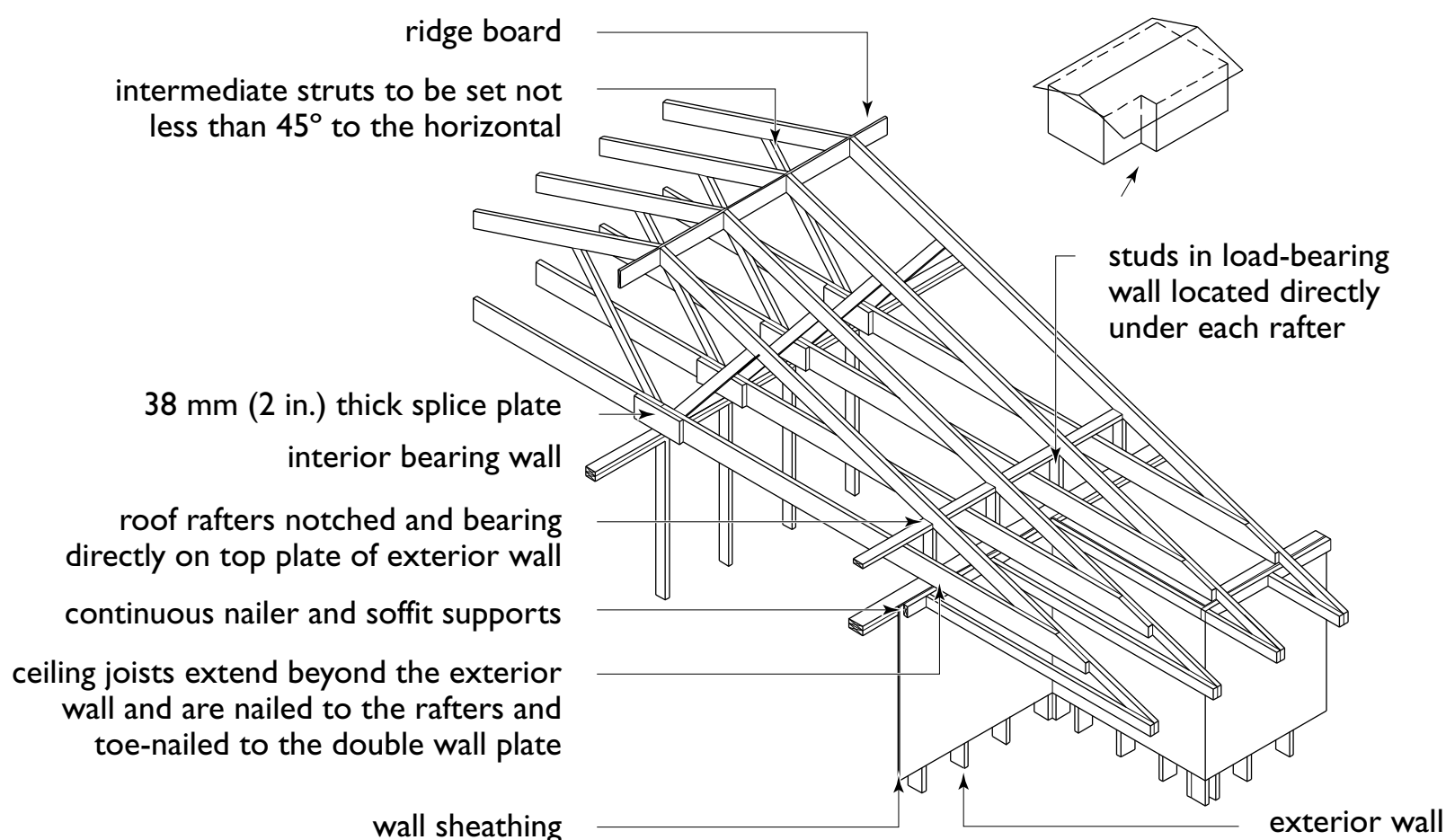
- directly on the wall plates (see Figure 83);
- on a rafter plate nailed to the top of the ceiling joists (see Figure 84); or
- on a load-bearing wall supported on the exterior wall plate (Figure 86).

The last method is used where a portion of the outside wall is inset. The ceiling joists extend beyond the exterior wall and are nailed to the side of the rafters. This provides lateral support for the bearing wall and stability for the ends of the rafters.

A ridge board (see Figure 83) or a ridge beam (see Figure 84) is used to ensure a level ridgeline and to facilitate placement of the rafters. Erect rafters in pairs and nail them to the ridge board or beam. Toe-nail the lower ends to the wall plate. Locate each pair of rafters directly opposite each other or offset at the ridge by not more than their thickness. Offsetting is required to maintain vertical alignment of the rafters when the lower ends are tied to ceiling joists that have been lapped (rather than butted together) at the centre load-bearing wall (see Figure 84).

A roof that slopes less than 1:3 should be vertically supported at the peak. This can be accomplished by providing a 38 × 140 mm (2 × 6 in. nominal) ridge beam supported at 1.2 m (4 ft.) intervals by 38 × 89 mm (2 × 4 in. nominal) vertical struts (see Figure 84). A load-bearing wall may be used instead of the ridge beam. Since the ridge support does not result in outward thrust of the roof members, continuous ties between the lower ends of

86 Rafter heel supported on load-bearing wall



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opposing rafters are not necessary. Ridge beams are also required for steeper pitched roofs when the outside ends of the rafters cannot be tied together to resist thrust.

Intermediate support is generally provided between the ridge and exterior walls to reduce the span of the rafters. This reduces the size of the rafters that is required, as the span is measured from this intermediate point to the ridge or eave support.

For rafters in roofs that slope 1:3 or more, intermediate support is usually provided by 38×89 mm (2×4 in. nominal) collar ties nailed to the side of each pair of rafters. Since these ties are in compression and subject to buckling, ties more than 2.4 m (8 ft.) long should be supported against lateral deflection. This can be done by nailing a 19×89 mm (1×4 in. nominal) continuous member at right angles to the collar ties near their centre with three, 76 mm (3 in.) nails at each end (see [Figure 83](#)).

Intermediate support for rafters in roofs that slope less than 1:3 is usually provided by a dwarf-bearing wall (see [Figure 84](#)) built in the same way as a load-bearing partition, except that a single top plate may be used where the rafters are positioned directly over the studs.

Struts may also be used as intermediate supports for rafters in pitched roofs. A 38×89 mm (2×4 in. nominal) strut (see [Figure 86](#)) is nailed

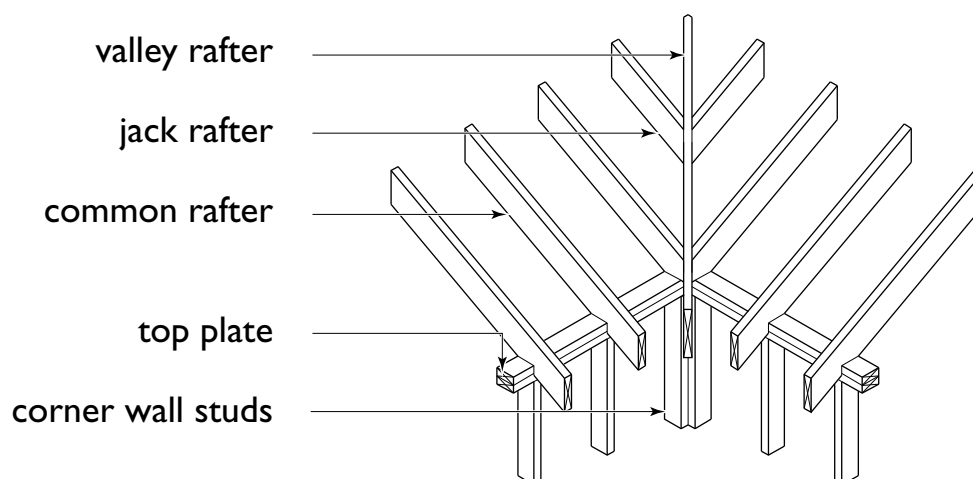
to the side of each rafter and supported on a load-bearing partition. The angle of any struts should not be less than 45° from the horizontal.

Rafters at right angles to the ceiling joists may be supported at an intermediate point by a dwarf wall sitting on a beam placed between the ceiling joists. The underside of the beam is raised at least 25 mm (1 in.) above the ceiling finish by blocks inserted under the ends of the beam at the exterior walls and intermediate load-bearing partition. The space prevents the beam from damaging the ceiling finish when deflected by the roof loads.

A beam may also be used as intermediate support for hip and valley rafters. A roof strut is used to transfer the load from the hip or valley rafter to the beam. Where intermediate support is required for a few rafters in the end section on a hip roof, a simple support can be provided by a beam (sometimes called a “strongback”) consisting of two, 38×89 mm (2×4 in. nominal) members nailed together, set on edge and nailed to the bottom of the rafters. This beam is supported at points along its length by 38×89 mm (2×4 in. nominal) struts radiating from a common point of support on the intermediate load-bearing wall. The ends are cut to fit the selected angle and securely nailed in place.

Hip and valley rafters should be about 50 mm (2 in.) deeper than common rafters ([Figures 83, 85 and 87](#)). This additional depth provides full

87 Framing at a valley



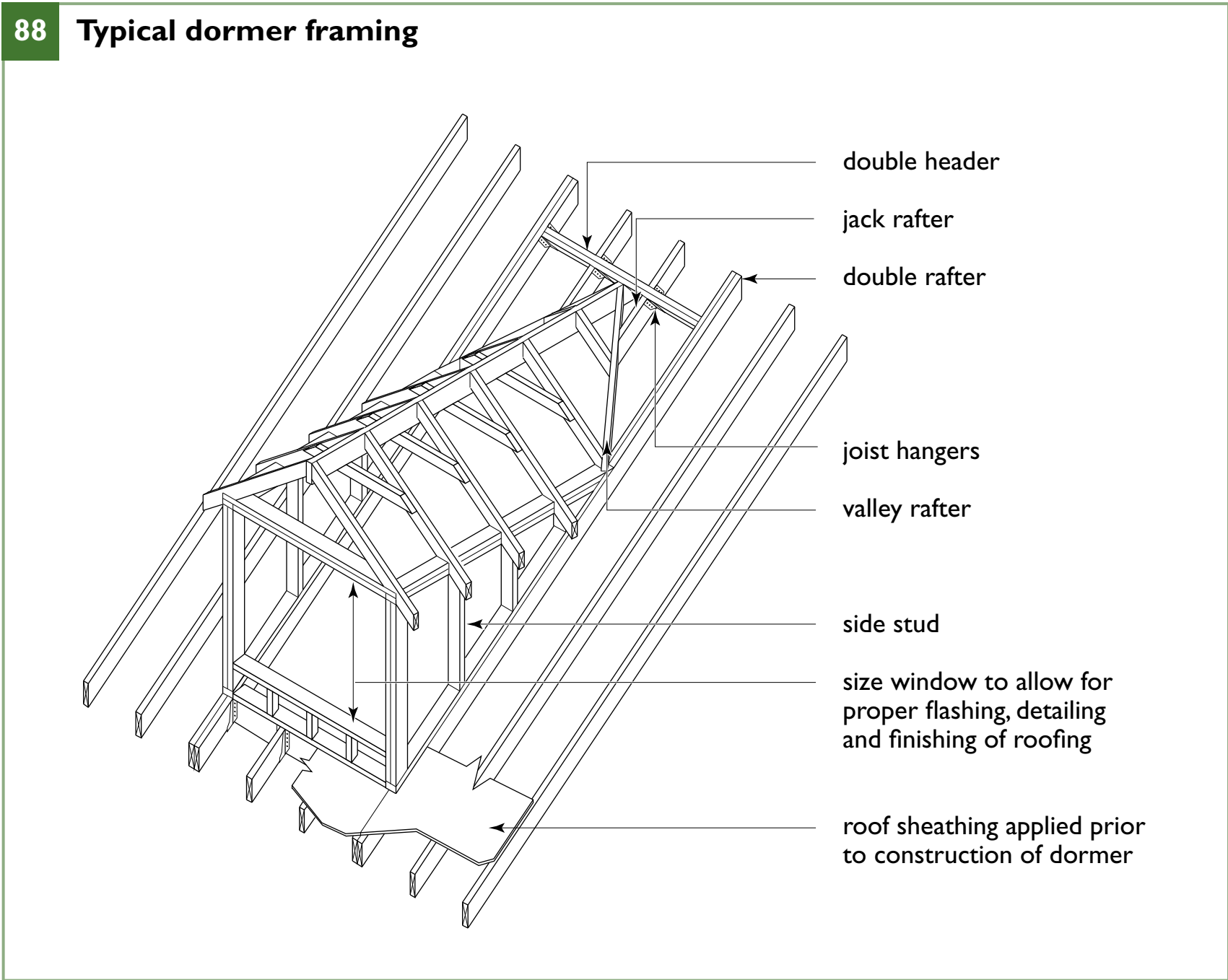
contact with the angle-cut of the jack rafters. In hip roofs, the jack rafters are nailed to the hip rafters and wall plate. Where a valley occurs, nail the jack rafters to the valley rafter and ridge.

Dormers are framed so that the rafters at each side are doubled to support the side studs and valley rafters. The top ends of the valley rafters are supported by a header spanning between the doubled rafters (Figure 88). The most common method of construction is to install the roof sheathing before the dormer is framed, and then cut the sheathing flush with the framing members

around the opening. A bottom plate added on top of the sheathing supports the side studs enclosing the dormer and also serves as a nailing base for the wall sheathing.

Gable-End Framing and Projections

After roof framing members have been installed, the gable-end studs are nailed in place. Studs in unfinished attics can be placed with the wide face parallel to the wall. Cut the ends of the studs to fit the angle of the rafter and toenail them to the wall plate and to the underside of the rafter with at



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least four 63 mm (2½ in.) nails at each end (Figure 89). The framing of the roof projections at the gable ends is shown in Figures 89 and 90.

Roofs that project less than 300 mm (12 in.) over the gable-end wall usually terminate with a framing member called a rake rafter

89

Wide projection at gable end supported by “lookout” rafters

ridge beam

rafter

top plate of gable end wall

blocking between lookouts to support roof sheathing and inner edge of soffit covering

lookouts at 600 mm (24 in.) o.c.

ceiling joist

top plate

header or rough fascia

fascia board at eave and gable end

corner studs

lookout cut away to show top plate

90

Narrow projection at gable end

blocking spaced at 600 mm (24 in.) o.c. supporting soffit covering

ridge beam

rafter

19 mm (¾ in.) nailing strip to support edge of soffit

ceiling joist

roof sheathing

gable wall sheathing taken up to underside of roof sheathing

top plate

studs

rake rafter supported by ridge beam, rafter header, blocking and sheathing

corner studs

rough fascia supports heel of rake rafter

(see Figure 90). A 19 mm ($\frac{3}{4}$ in. nominal) nailing strip is fastened to the rafter located above the gable end wall. Blocking spaced at 600 mm (24 in.) on centre is used to support the soffit covering. This blocking is toenailed to the nailing strip and end-nailed to the rake rafter. The soffit covering is then installed and nailed to these supports and a fascia board is added.

Gable-end projections extending more than 300 mm (12 in.) beyond the wall should be supported by framing members called “lookouts” (see Figure 89). Gable-end studs are placed with the narrow face parallel to the sheathing, and a top wall plate is added. The lookout members, usually the same size as the rafters, are spaced at 600 mm (24 in.) on centre.

The ends are supported by end-nailing to the first rafter and inboard of the gable end to the top plate of the gable. Blocking is then fitted between the lookouts in line with the wall line to support the roof sheathing and inner edge of the soffit covering. The soffit covering is nailed to these supports, and a fascia is added. The length of lookout members should be at least twice the width of the roof overhang. A double rafter is used to support the inner ends of the lookout members when they project into the roof more than one-and-one-half rafter spaces.

Low-Slope Roofs

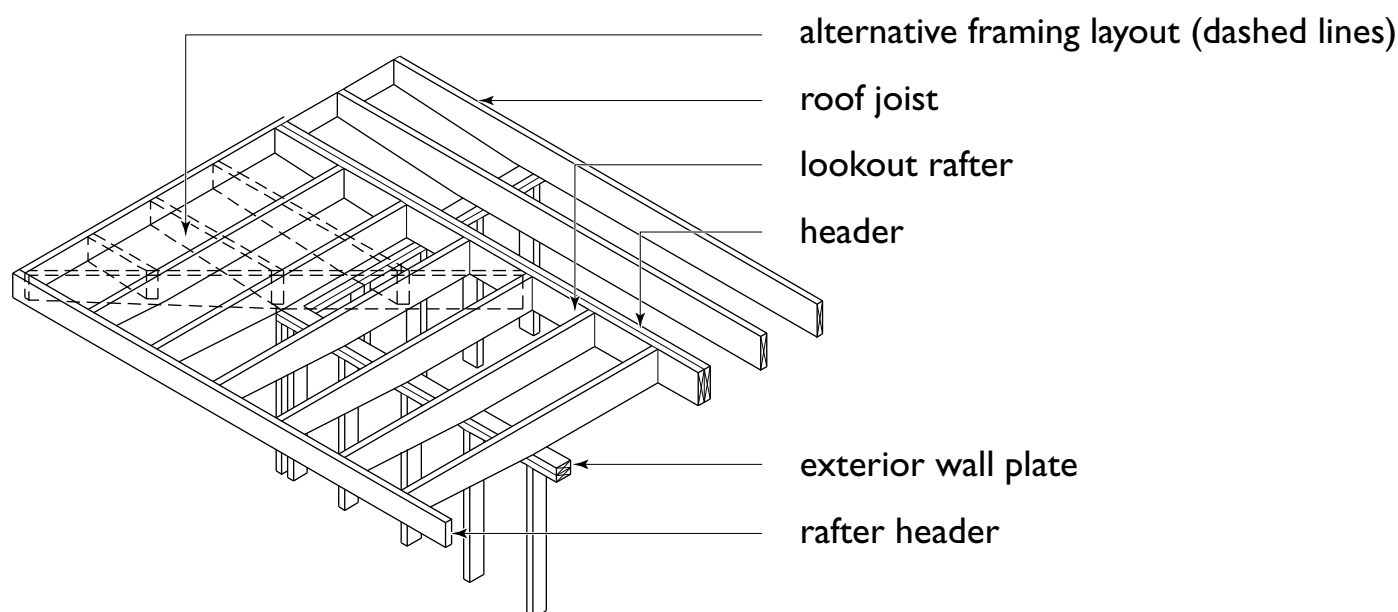
Low-slope roofs must be constructed more water-tight than pitched roofs because drainage is poorer. They sometimes support decks or terraces or occasional traffic and therefore must be protected with durable, wear-resistant materials.

Where rafters also serve as ceiling joists, the term “roof joist” is used. Their size is determined by both roof and ceiling loads. (See Tables 29 and 30 on pp. 295-297) Rafters chosen for structural adequacy may not, however, provide adequate depth for insulation and ventilation of the roof space. In such cases, wider lumber or engineered wood products should be used.

Roof joists for low-slope roofs are usually laid level or nearly level, with roof sheathing and a roof covering on top. The underside of the roof joists supports the ceiling. A slope of at least 1:50 is required for drainage. Slope the joists by adding a ledger strip at the top of the bearing wall or by adding a tapered strip to the top of the joists.

Lookout rafters are usually used when an overhang is provided for a low-slope roof (Figure 91). If the lookout rafters project into

91 Typical construction of a low-slope roof with overhang



the ceiling area more than one and one-half joist spaces, two roof joists are nailed together to form a header. One outside rafter header is then added and end-nailed to the lookout rafters and roof joists. This serves as a nailing support for the roof sheathing, fascia board and soffit covering.

ROOF SPACE VENTILATION

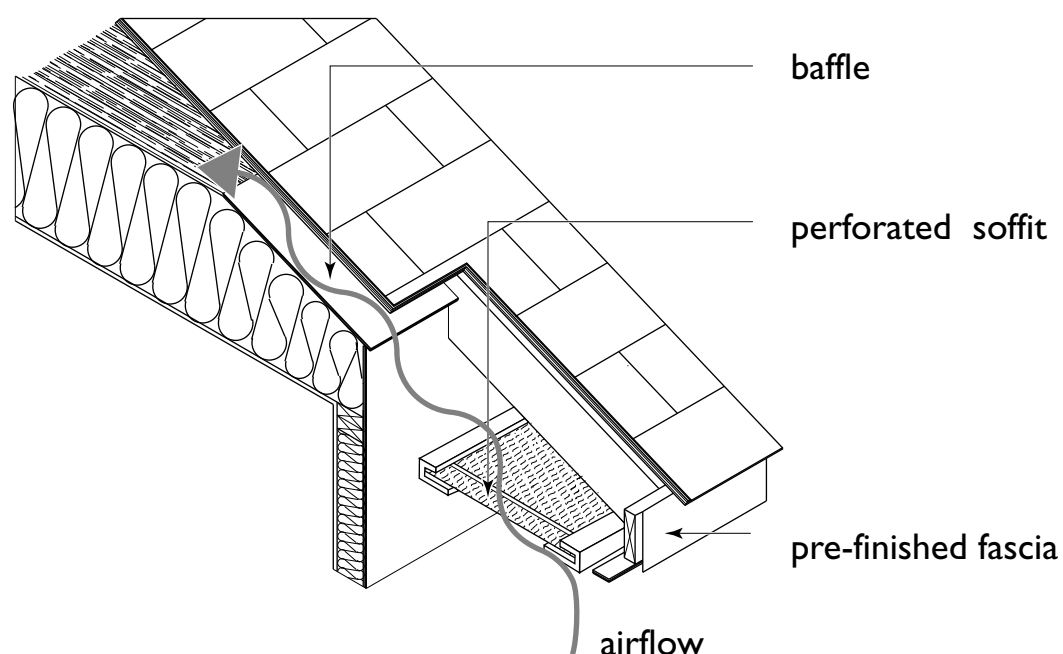
Always provide adequate ventilation of the roof space above the insulation. Accumulated water vapour in attic spaces and under low-slope roofs could condense during cold weather in sufficient quantity to cause damage. Since most types of roof membranes are highly resistant to vapour transmission, vapour does not pass through them and ventilation is needed to remove vapour that enters the roof space.

During cold weather, heat lost through the ceiling insulation combined with sun exposure may provide enough heat to melt the snow on a roof but not on the projecting eaves. Water from the melting snow can freeze and form ice dams

at the eavestrough and roof overhang, which are exposed to outdoor temperatures on all sides and are therefore colder than the ceiling. This may cause water to back up at the eaves, penetrate through the roof and leak into the walls and ceilings (see [Figure 100](#)). Adequate airtightening in combination with insulation and ventilation beneath the roof sheathing will keep attic temperatures low during winter and help prevent snow on the roof from melting. A space above the insulation between the exterior walls and the underside of the roof sheathing allows ventilation from the eaves to the attic space. When no air ventilation baffles are used, the clearance must be at least 63 mm (2½ in.). If baffles are used, the clearance can be reduced to 25 mm (1 in.) (see [Figure 24](#) on p.56).

Roofs with slopes of 1:6 and greater are required to have an unobstructed vent area at least $\frac{1}{300}$ of the insulated ceiling area, measured on a horizontal plane. For roofs with slopes less than 1:6, the area must be at least $\frac{1}{150}$ of the ceiling area. Roof-, gable- or eave-type vents can be used (see [Figures 92 and 93](#)). Vents must be uniformly

92 Soffit roof ventilation



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distributed with at least 25 per cent of the vent area located at the bottom of the space and at least 25 per cent at the top.

If batt insulation is placed between the roof-ceiling joists in a low-slope roof, a ventilated space of at least 63 mm (2½ in.) must be provided between the top of the insulation and the underside of the roof sheathing. This can be achieved by placing 38 × 64 mm (2 × 3 in. nominal) cross-purlins over the top of, and at right angles to, the roof-ceiling joists. The cross-purlins may be shimmed to provide the required roof slope (see [Figure 25](#) on p. 57). If rigid

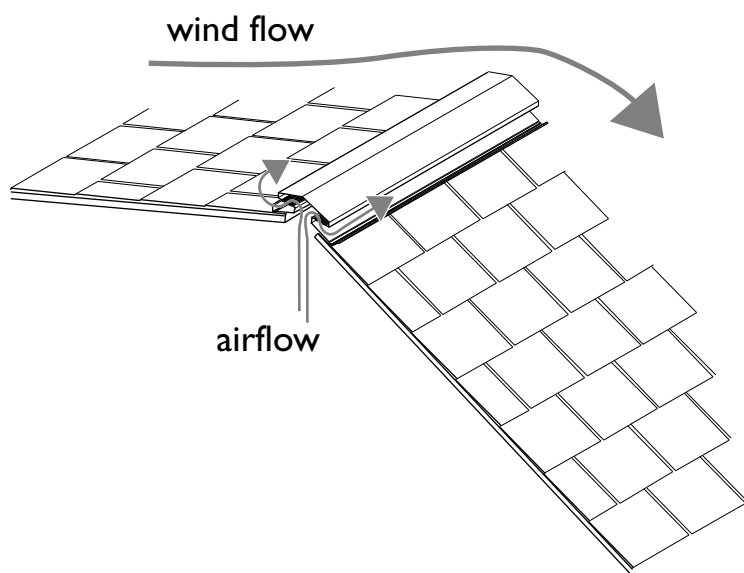
insulation is installed on top of the roof sheathing (and below the roof covering), the space above the ceiling is not ventilated.

Vents must not allow the entry of rain, snow or insects. If a vent is a potential source of moisture entry, a baffle is required to reduce air flow. Corrosion-resistant metal or plastic materials should be used for vents and to screen ventilator openings.

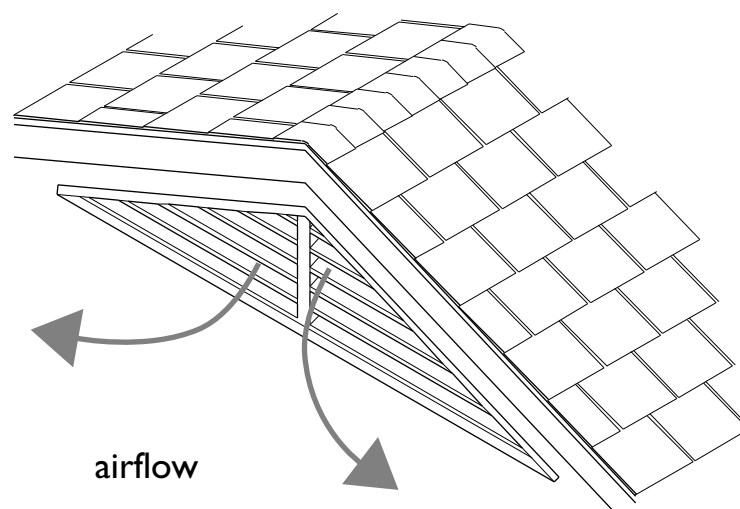
Except for very small attic spaces, attics require an access hatch. The area of the hatch must be at least 0.32 m² (3.4 sq. ft.) with no dimension less than 500 mm (20 in.).

93 High-level roof ventilation

Ridge vent



Gable vent



CHOOSING THE SIZE AND SPACING OF CEILING JOISTS

Problem

Select the minimum ceiling joist that is acceptable for the following conditions:

Conditions

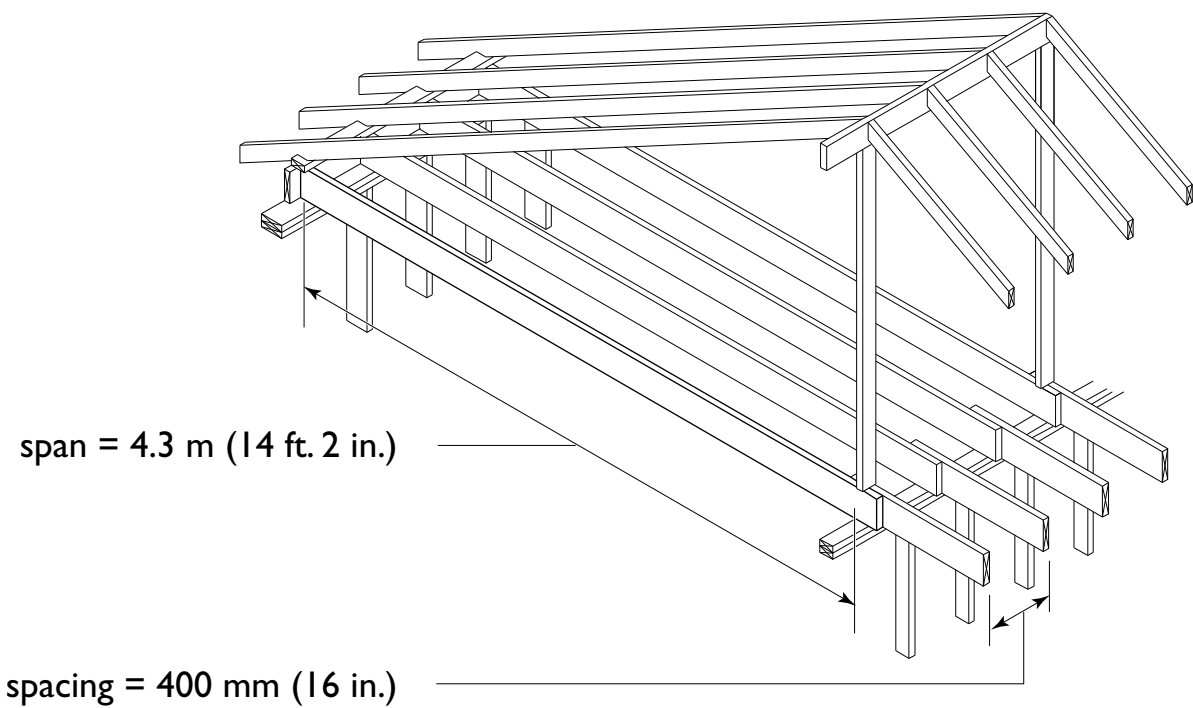
- Attic is inaccessible and is not used for storage.
- Ceiling supports the insulation and drywall interior finish.
- Span is 4.3 m (14 ft. 2 in.).
- Ceiling joist spacing is 400 mm (16 in.).
- Species group and grade specified is SPF No. 2 and better.

Selection

Use Table 33 (p. 303).
For this application select 38 × 140 mm (2 × 6 in.) ceiling joist.

Note: In accordance with the National Building Code, the metric spacing of wood framing members is expressed as a “soft” conversions from the actual imperial dimensions. For example, spacing of 12, 16 and 24 inches on centre is converted to 300, 400 and 600 mm. on centre, respectively. In order to suit the imperial sizes of common 1 220 × 2 440 mm (4 × 8 ft.) panel products, such as gypsum board, OSB and plywood, the actual spacing of framing members is commonly adjusted to 305, 406 and 610 mm on centre, respectively.

94 Ceiling joists



CHOOSING THE SIZE AND SPACING OF ROOF RAFTERS

Problem

Select the smallest dimension rafter that is able to span the roof as follows.

Conditions

- Building is located in Ottawa.
- Specified roof snow load is 1.72 kPa (36 lbs/sq. ft.).
- Roof slope is 1:3.
- Rafter span is 4.7 m (15 ft. 6 in.).
- Species group and grade specified is SPF No. 2 and better.
- Shingle roofing to be used.

- Lower ends of the rafter are restrained.

Selection

Use Table 31 (p. 299).

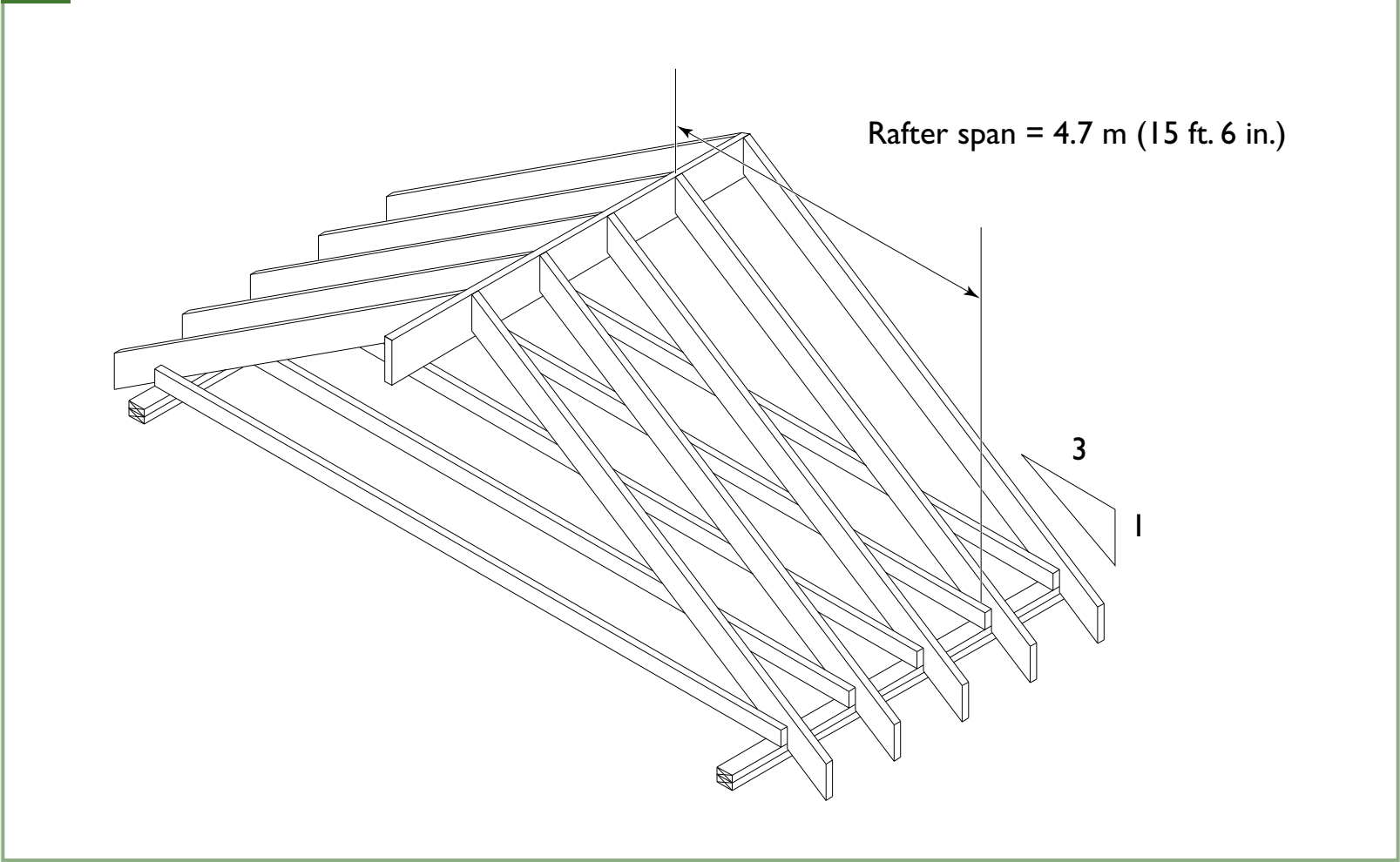
Acceptable rafters include:

38 × 184 mm (2 × 8 in.) at 300 mm (12 in.) on centre.

38 × 235 mm (2 × 10 in.) at 600 mm (24 in.) on centre.

Note: The rafters must also be deep enough to accommodate the required level of insulation.

95 Roof rafters



CHOOSING THE SIZE AND SPACING OF ROOF JOISTS

Problem

Select the roof joists that are able to span the roof as follows.

Conditions

- Building is located in Ottawa.
- Specified roof snow load is 1.72 kPa (36 lbs/sq. ft.).
- Roof slope is 1 in 3.
- Roof joist span is 4.19 m (13 ft. 9 in.).
- Species group and grade specified is SPF No. 2 and better.

- Shingle roofing to be used.
- Lower ends of the roof joists are restrained.

Selection

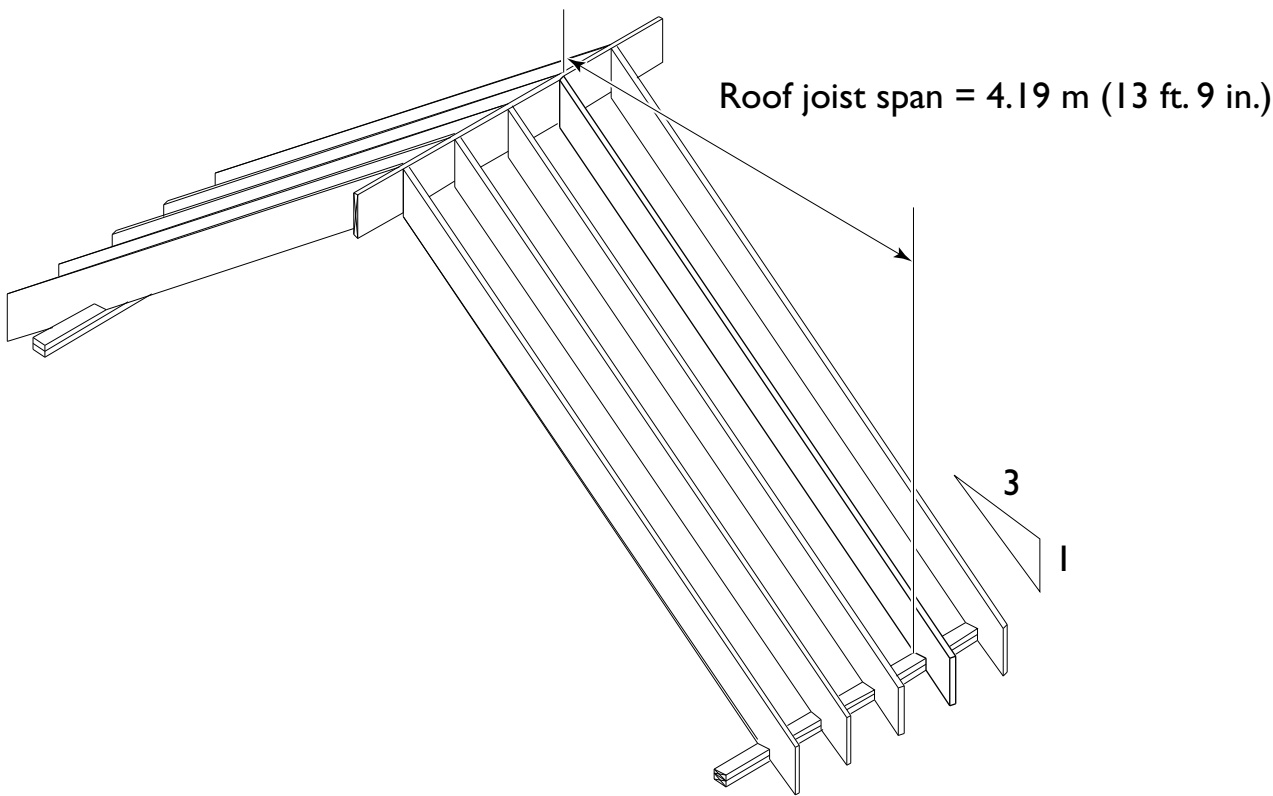
Use Table 29 (p. 295).

Acceptable roof joists include:

- 38 × 235 mm (2 × 10 in.) at 400 mm (16 in.) on centre.
- 38 × 286 mm (2 × 12 in.) at 600 mm (24 in.) on centre.

Note: The roof joists must also be deep enough to accommodate the required level of insulation.

96 Roof joists



SUSTAINABLE HOUSING INSIGHT

Energy Efficiency

- Use raised heel trusses or rafters/roof joists that are deep enough to contain high levels of insulation over exterior walls.
- Consider using raised heel trusses that are deep enough at the eaves to accommodate higher levels of insulation than required by the building code.
- Consider increasing the depth of roof framing members or reducing the spacing between them to accommodate future solar thermal and photovoltaic panels. Consult a structural engineer.

Resource Efficiency

Prefabricated roof trusses are widely used for residential roof framing. Trusses use an efficient arrangement of select lumber and metal plates to provide long spans and reduce the amount of wood required as compared with framing with lumber rafters and joists. They provide the following advantages:

- Designed to suit local climatic conditions.
- Fast to erect leading to earlier close-in.
- Ample space for insulation and ventilation.
- Available to suit all possible roof shapes.
- Eliminate cut-offs at the site.

Affordability

When planned in advance, attic space can be converted to high-quality living space. This is economical because it provides additional space without increasing the building footprint and makes use of an existing roof. Consider these factors when planning attic rooms:

- Plan stair locations so access can be extended to the attic space in the future.
- Size ceiling joists as floor joists or design the attic trusses or rafters for future habitable space. Use ceiling insulation that may be moved to the roof assembly during conversion.
- Ensure that the roof assembly will accommodate sufficient insulation with a ventilation air space above it.
- Steeper roof slopes shed snow and rain more effectively and enclose more attic volume than lower slopes.
- Rough in or plan for future services such as electricity, plumbing, heating, ventilation, cable and telephone.

CHAPTER 12

Roof Sheathing and Coverings

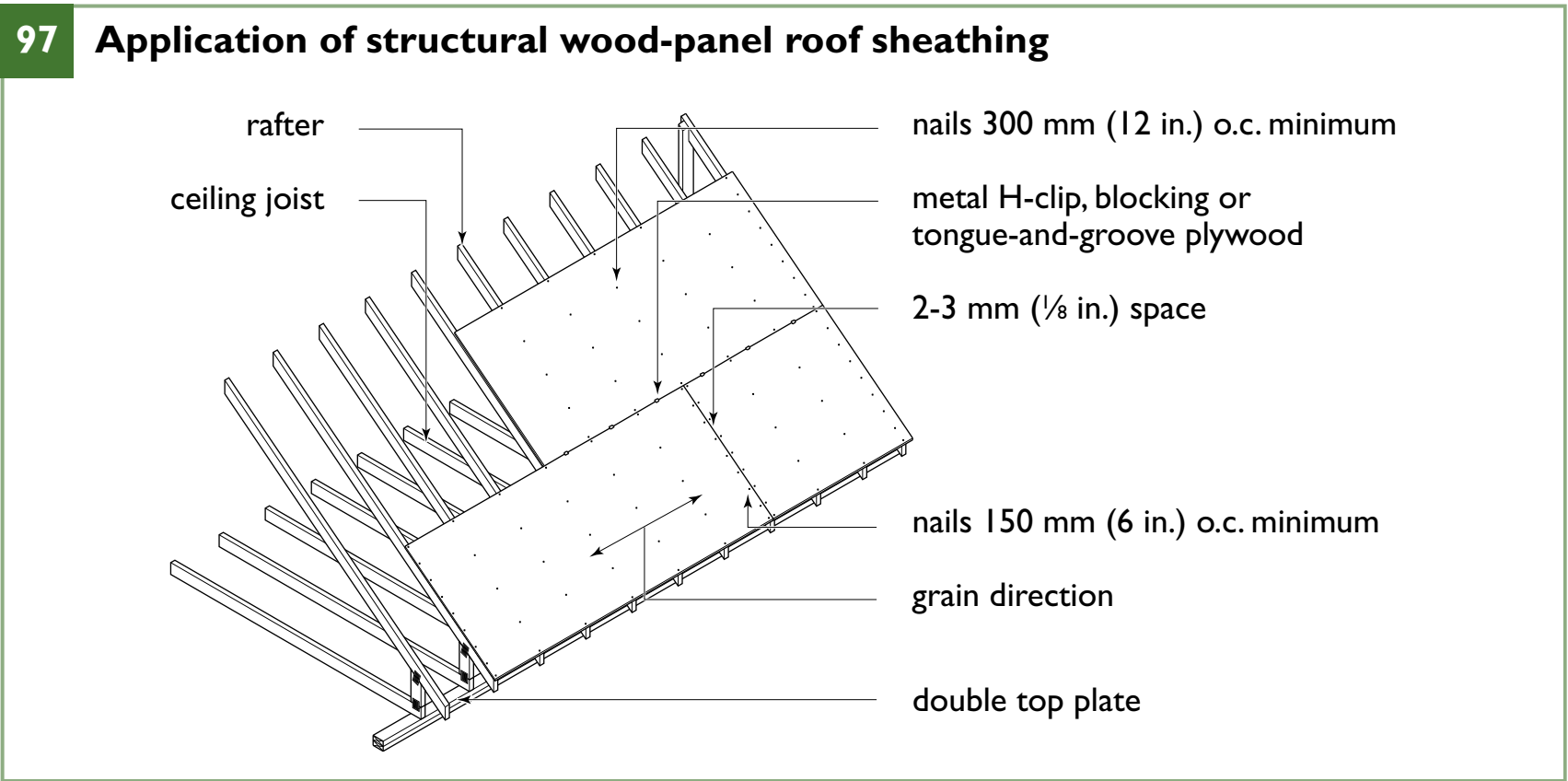


ROOF SHEATHING

Sheathing-grade plywood or OSB or lumber planks are applied over roof trusses, joists or rafters to provide a nailing base for the roof covering and lateral bracing of the roof framing.

Installing Roof Sheathing

When plywood or OSB is used for roof sheathing, sheets are laid with the face grain at right angles to the framing ([Figure 97](#)) and with the joints offset to obtain a good



tie across the roof framing. Leave a 2 to 3 mm ($\frac{1}{8}$ in.) gap between panels to prevent buckling should expansion in the sheathing occur.

The required thicknesses of the plywood, OSB or other roof sheathing are shown in Table 35 (p. 305). Sheathing thickness depends on the spacing of the rafters, roof joists or trusses and whether the edges of the sheets are supported. To prevent damage to the roof covering when thinner panels are used, support the joints with 38×38 mm (2 $\times$ 2 in. nominal) blocking nailed securely between the roof framing members, or by metal H-clips inserted between sheets. H-clips are widely used because installation is simple and economical.

It is good practice to stagger the location of fasteners at the edges of adjoining panels. Staple fasteners for 9.5 mm ($\frac{3}{8}$ in.) roof sheathing must be at least 1.6 mm thick, 38.1 mm long with a 9.5 mm crown ($\frac{1}{16}$ in. thick, $1\frac{1}{2}$ in. long with a $\frac{3}{8}$ in. crown), and be driven with the crown parallel to the framing (Table 38 on p. 307). Thicker roof sheathing (Table 22 on

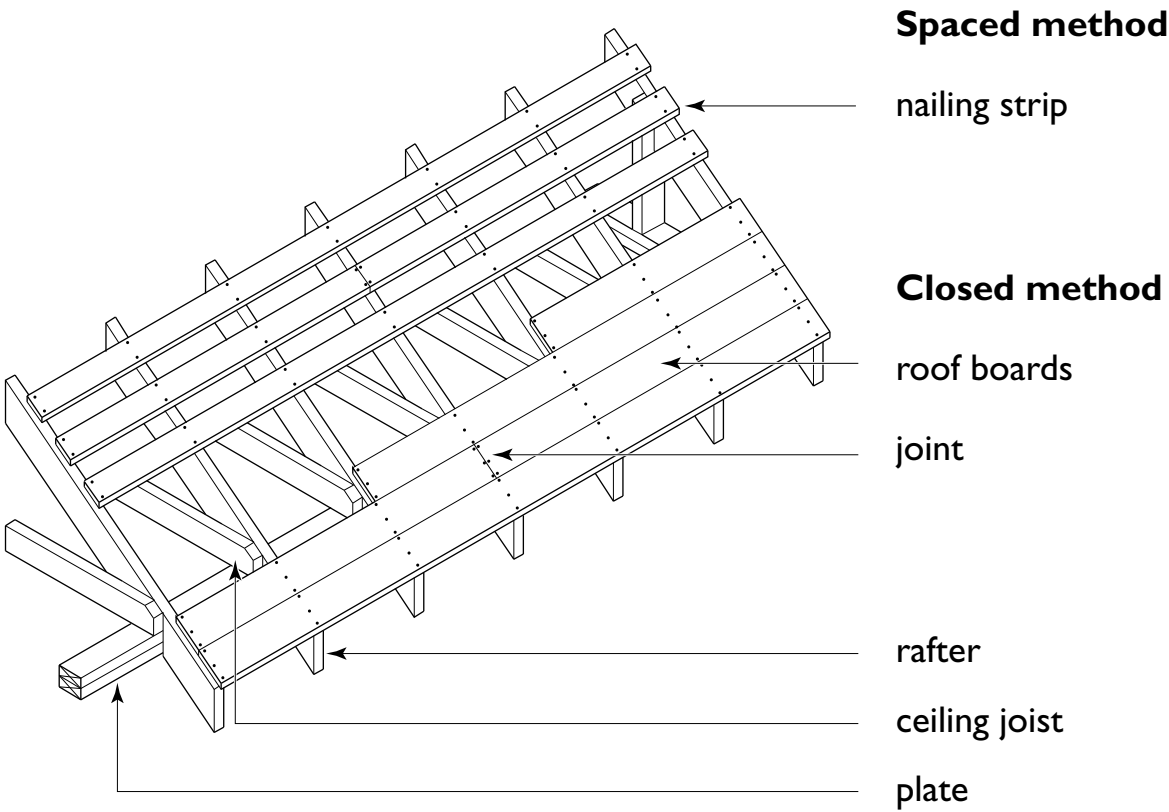
p. 287)—no less than 15.5 mm ($\frac{5}{8}$ in.)—is required for built-up roofing on a low-slope roof to be used as a walking deck.

Lumber sheathing must be laid closed, or edge to edge (Figure 98) with all ends supported and joints staggered. For exposure to high wind or earthquake forces, lumber sheathing must be laid diagonally or be engineered. The boards are usually 19 mm ($\frac{3}{4}$ in. nominal) thick, but this thickness may be reduced to 17 mm ($\frac{11}{16}$ in.) where supports are spaced at 400 mm nominal (16 in. actual) or less on centre.

Boards 184 mm (8 in. nominal) or less in width are nailed to the framing members with at least two, 51 mm (2 in.) nails per bearing. Those wider than 184 mm (8 in. nominal) should be nailed with at least three, 51 mm (2 in.) nails per bearing. For areas with high wind loads, the length of the nails must be increased to 63 mm ($2\frac{1}{2}$ in.).

Boards wider than 286 mm (12 in. nominal) should not be used for roof sheathing. For a wood shingle roof, the roof boards may be spaced

98 Installation of wood-board roof sheathing



Note: For areas exposed to high wind and earthquake forces, lumber roof sheathing must be installed diagonally.

the same distance apart on centre as the shingle exposure. This method (Figure 98), commonly used in damp climates, permits air movement and allows shingles to dry between wettings, reducing the possibility of decay.

Roof Sheathing Details

Where openings occur in the roof structure for interior chimneys, the roof sheathing and framing members should have a clearance of 50 mm (2 in.) on all sides from the finished masonry, or metal chimney for fire safety (Figure 99). This clearance may be reduced to 12 mm (½ in.) for exterior masonry chimneys. Roof sheathing should be securely nailed to the rafters and headers around the opening.

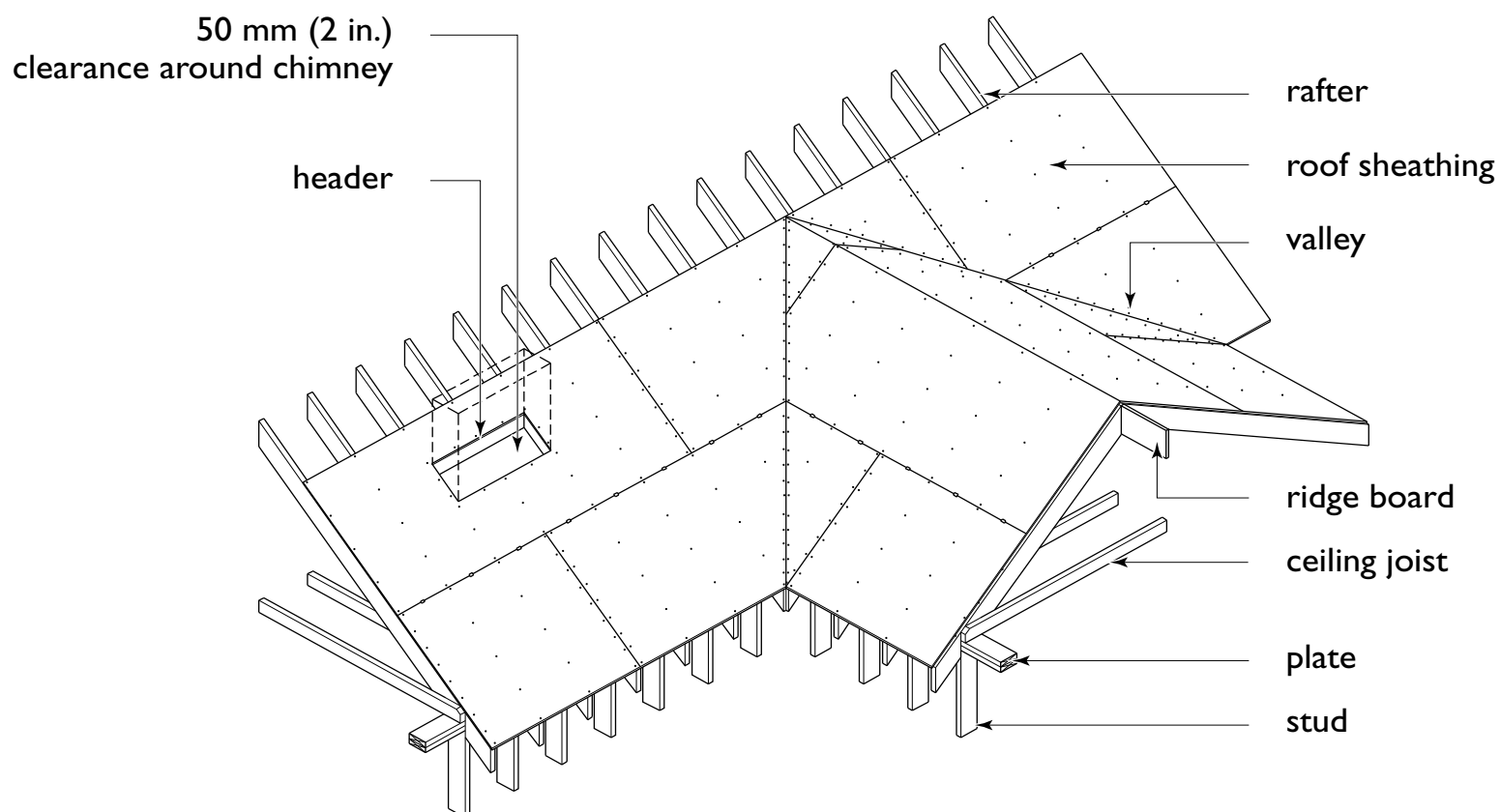
Fit sheathing at valleys and hips to give a tight joint and securely nail it to the valley or hip rafter (Figure 99). This will provide a solid, smooth base for flashing (see Chapter 14).

EAVE PROTECTION

Eave protection is achieved by installing a membrane along the edge of the roof to prevent water from entering the roof. It is not required over unheated garages and porches, on roofs with a slope of 1:1.5 or greater or in regions with 3,500 or fewer heating degree-days. Eave protection should extend at least 900 mm (36 in.) up the roof to a point at least 300 mm (12 in.) inside the inner face of the exterior wall. Eave protection is installed under the shingle starter course and is most often a self-adhering bituminous or roll roofing membrane.

During cold weather, heat loss through the ceiling insulation combined with sun exposure may provide enough heat to melt the snow on a roof but not on the projecting eaves. Water from the melting snow can freeze and form ice dams at the eavestrough and roof

99 Roof sheathing detail at a valley and at a chimney opening

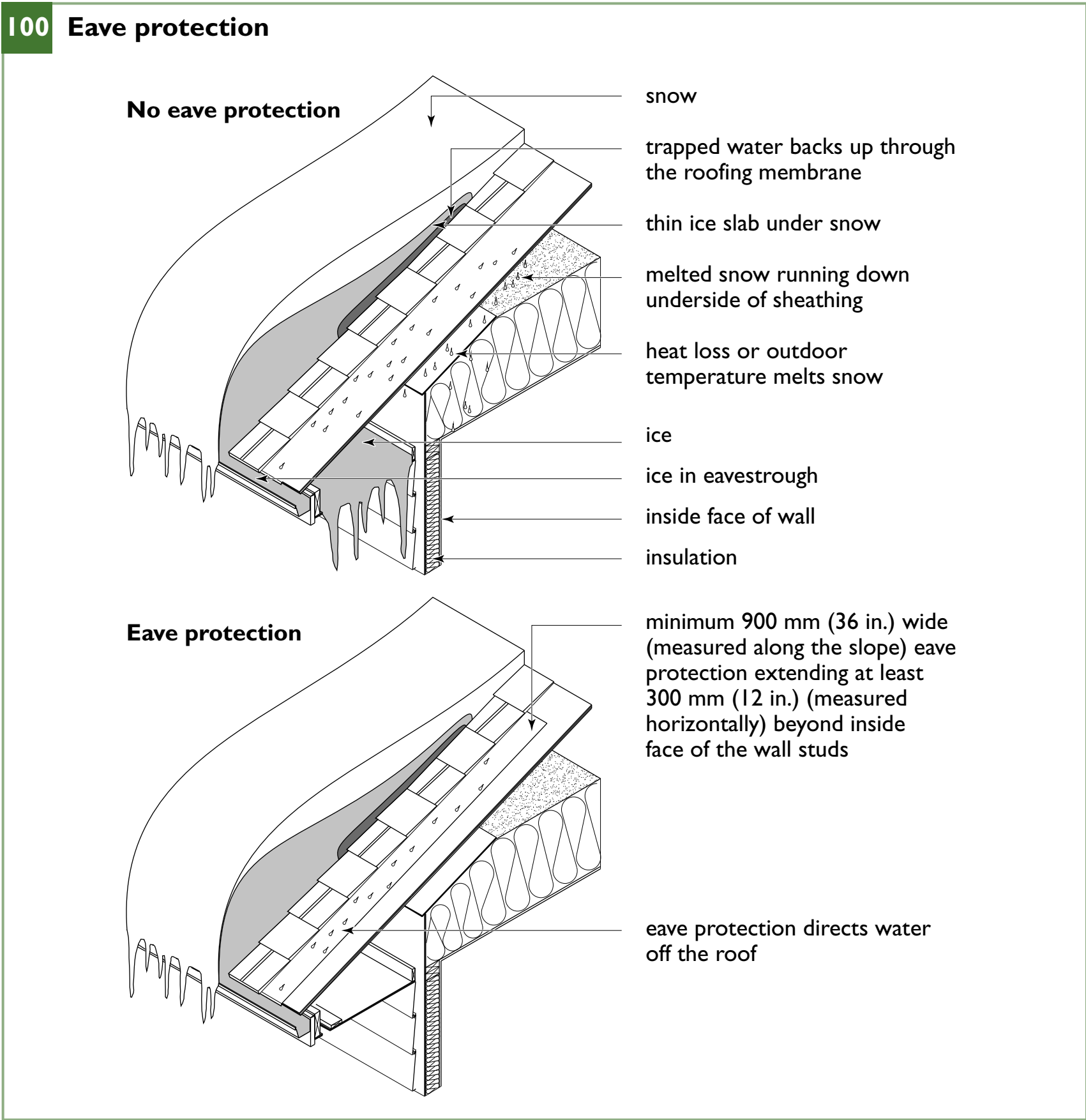


overhang. As temperatures fluctuate, the melted snow freezes, causing the ice dam to progressively grow larger (Figure 100).

When a thaw occurs, this ridge of ice causes a dam that prevents melt water from escaping down the roof. The trapped water backs up under shingles and could leak into the attic, ceiling and wall. To prevent ice damming from occurring, do pay attention to air-sealing at the ceiling, the

ceiling/wall junction and at all penetrations through the ceiling air barrier during construction. Provide sufficient ceiling insulation to comply with building code requirements for your location, and ensure that the attic space is vented to the exterior at the eaves.

Skylights should be situated away from the areas where ice damming is likely. In addition, eave protection consisting of either type S



(smooth surface) or type M (mineral-surfaced) roll roofing should be installed over the sheathing, laid with the joints lapped at least 100 mm (4 in.) and cemented together. Alternatively, a peel-and-stick bituminous membrane may be used. Methods of flashing shingled roofs at valleys, chimneys and intersecting walls are described in Chapter 14.

ROOF COVERINGS

The roof covering should be installed as soon as the sheathing is in place to keep the building dry and to provide durable, water-resistant protection.

The choice of roofing materials may be influenced by cost, appearance, local code requirements or local preferences. Asphalt shingles are commonly used on pitched roofs. Prefinished metal, galvanized steel or aluminum roofing is also common in some regions. At normal roof pitches, metal roofs will generally shed snow, a desirable characteristic in heavy snowfall areas. Roll roofing, wood shingles, hand-split shakes, sheet metal and concrete or clay tile are also used. Built-up roofing with a gravel topping or cap sheet is usually used for low-slope roofs.

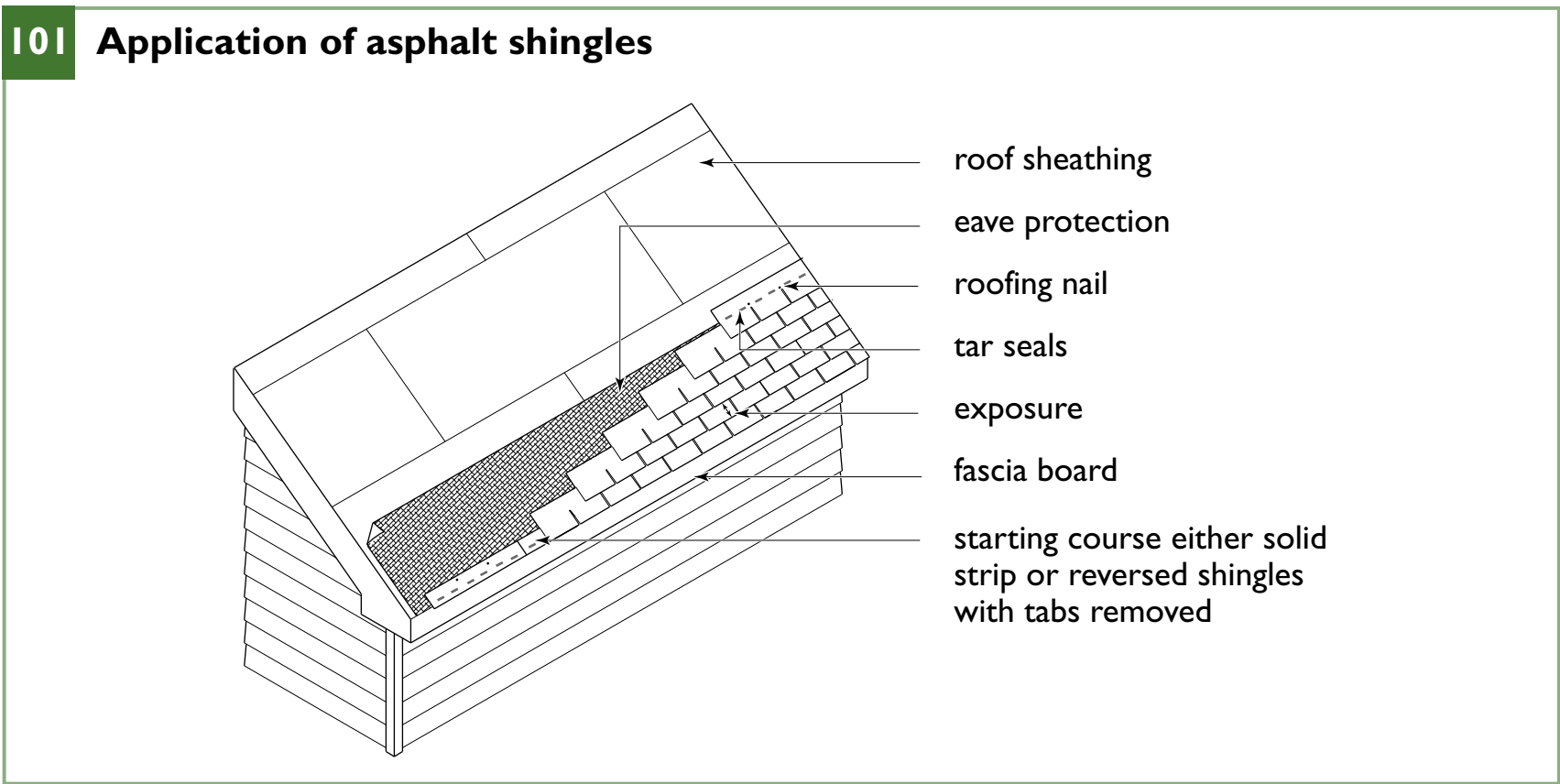
The minimum and maximum slopes for the different types of roof coverings are presented in Table 36 (p. 306). The minimum slope of roofs is 1:6 for asphalt shingles (using a low-slope application), 1:4 for wood shingles, and 1:3 for hand-split shakes and asphalt shingles (using normal application).

**Asphalt Shingles on Slopes
1:3 or Greater**

Asphalt strip shingles should be a minimum No. 210 grade. Square-butt strip shingles are usually 310 × 915 mm (12 × 36 in.) or 335 × 1 000 mm (13¼ × 39 ⅜ in.) in size, have three tabs and should be laid with 130 mm (5 in.) or 145 mm (5¾ in.) of their width exposed to the weather. Since there are approximately 21 to 26 strips in a bundle, a bundle will cover about 3 m² (32 sq. ft.).

Store bundles flat so that strips will not curl when the bundles are opened. Distribute the storage of shingles on roofs so that the load capacity of the framing is not exceeded.

The method for installing asphalt shingles is shown in **Figure 101**. Install the eave protection where required. Apply a starter



CHAPTER 12

Roof Sheathing and Coverings

strip comprised of shingles with their tabs removed along the eaves before the first course of shingles, so that it extends at least 12 mm (½ in.) beyond the eaves, rakes and fascia board to form a drip edge. Starter strips with metal drip edges can also be used.

This projection prevents water from backing up under the shingles by capillary action. Alternatively, a shingle strip laid with the tabs facing up the roof slope may be used for this purpose. Type M roll roofing may also be used as a starter strip and, when continued up the roof slope, will also serve as eave protection. Nail the starter strip along the bottom edge at 300 mm (12 in.) intervals. Lay the first course of shingles with the butt edge in line with the bottom of the starter strip.

The allowable exposure distance of shingles must be respected and depends on the roof slope and the type and length of shingle used.

Chalk lines can be used to align the shingle courses so that tabs and tab notches are in a straight line. Asphalt-impregnated building paper is often placed over the roof sheathing as a shingle underlay, and the lines printed on the underlay serve as guides to help align the shingles. Use four large-head roofing nails per strip that are long enough to penetrate at least 12 mm (½ in.) into the roof sheathing. Most asphalt shingles are self-sealing; a strip of adhesive on the underside of the tab affixes the shingle to the one beneath.

Interlocking and other special shingles should be laid according to the manufacturer's directions.

Asphalt Shingles on Low Slopes of 1:6 to 1:3

Care is required to ensure that the roof covering on a low slope roof is waterproof. Except for the first two courses, use three thicknesses of shingles on the entire roof including hips and

ridges. Maintain a shingle exposure of not more than one-third the full height of the shingle. Install starter strips as for higher-sloped roofs, but apply them over a continuous band of cement not less than 200 mm (8 in.) wide. The first course of shingles is then cemented to the strip with a continuous band of cement that is 100 mm (4 in.) wider than the shingle exposure. For example, a 250 mm (10 in.) band of cement should be used when the shingle exposure is 150 mm (6 in.). The succeeding courses of shingles are laid on a band of cement 50 mm (2 in.) wider than the shingle exposure, so that a 200 mm (8 in.) band is used when the exposure is 150 mm (6 in.).

To avoid defacing the exposed surface of the shingles with cement, the band should be located between 25 and 50 mm (1 and 2 in.) above the butt edge of each succeeding course of shingles. Staple or nail each shingle strip in place with four fasteners located above the exposed portion of the shingle. If cold-application cement is used, it should be applied at a rate of approximately 0.5 L/m² (1 gal./100 sq. ft.) of cemented area. Hot-application cement is applied at a rate of approximately 1 kg/m² (0.2 lb./sq. ft.) of cemented area. This technique is necessary only for slopes lower than 1:4 since there are special low-slope shingles of sufficient length to maintain the three thicknesses necessary at roof pitches of 1:4 or greater.

Wood Shingles

Western red and white cedar are the principal species of wood used for shingles because their heartwoods have high decay resistance and low shrinkage (they may also be preservative-treated). Other species are also used for shingles but should be preservative-treated. Wood shingles commonly for houses are No. 1 and No. 2 grade. The width of wood shingles varies between a maximum width of 350 mm (14 in.) and a minimum width of 75 mm (3 in.).

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Roof Sheathing and Coverings

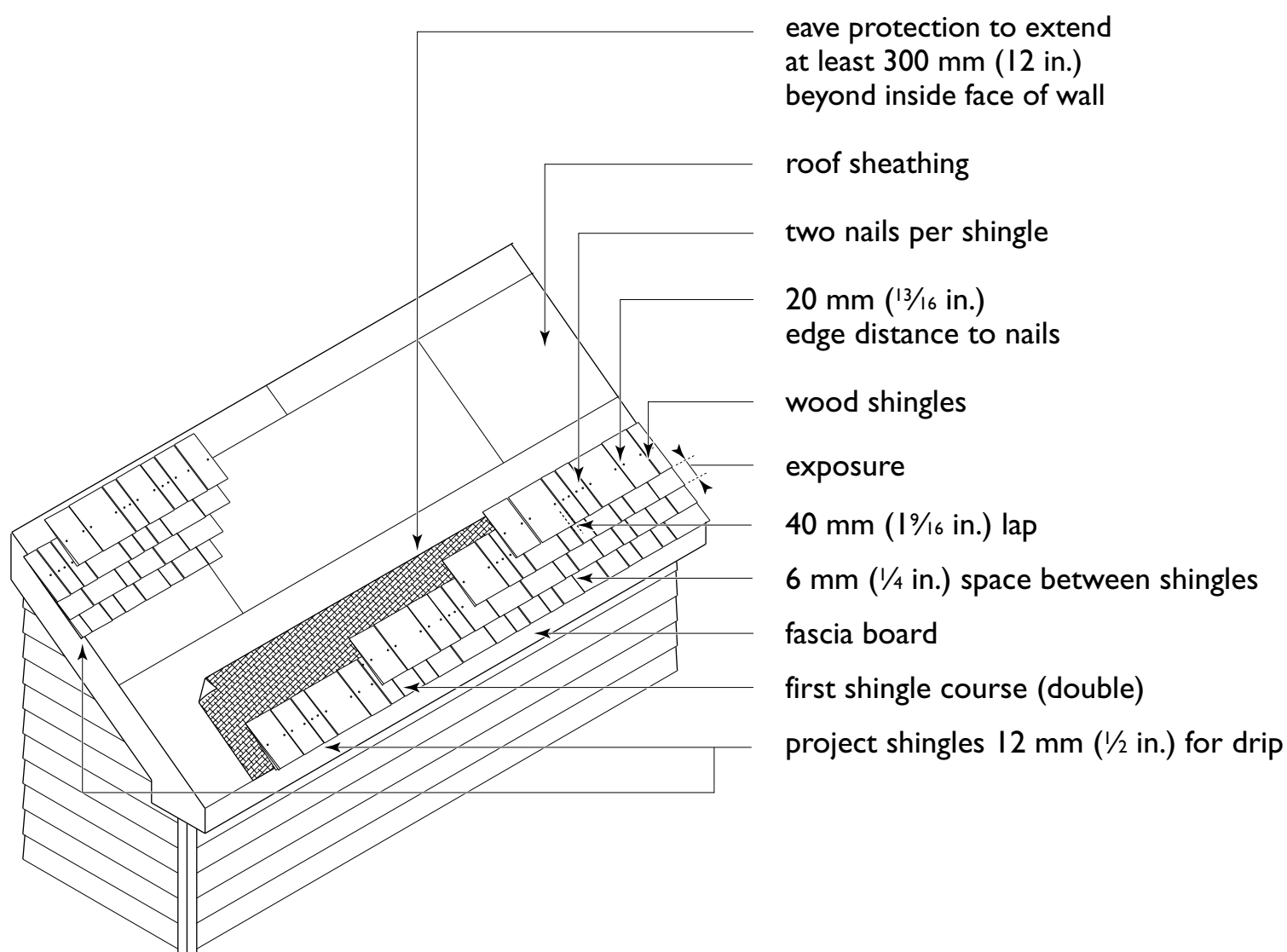
Figure 102 shows how to lay a wood shingle roof. As for asphalt shingles, underlay and roofing felt is not usually required, but eave protection is required.

Double lay the first shingle course with the upper shingles overlapping the joints in the course beneath and both rows extending about 25 mm (1 in.) beyond the fascia boards at the eaves. This will prevent water from backing up underneath the shingles. Place shingles 6 mm ($\frac{1}{4}$ in.) apart to allow for swelling. Offset the joints between shingles in one course at least 40 mm ($1\frac{5}{16}$ in.) from a joint in the

course below. Space the joints in succeeding courses so that the joint in one course is not in line with the joints in the two previous courses laid.

Use only two nails per shingle. The distance of the nails from the butt edge should be the shingle exposure plus 40 mm ($1\frac{5}{16}$ in.), with an edge distance of about 20 mm ($1\frac{3}{16}$ in.). For example, if the shingle exposure is to be 125 mm (5 in.), a nail should be 165 mm ($6\frac{5}{16}$ in.) from the butt edge of the shingle being nailed. Use hot-dip galvanized or other corrosion-resistant shingle nails.

102 Installation of wood shingles



Flat grain shingles wider than 200 mm (8 in.) are sometimes split and nailed as two shingles to avoid problems with cupping and warping.

Shakes

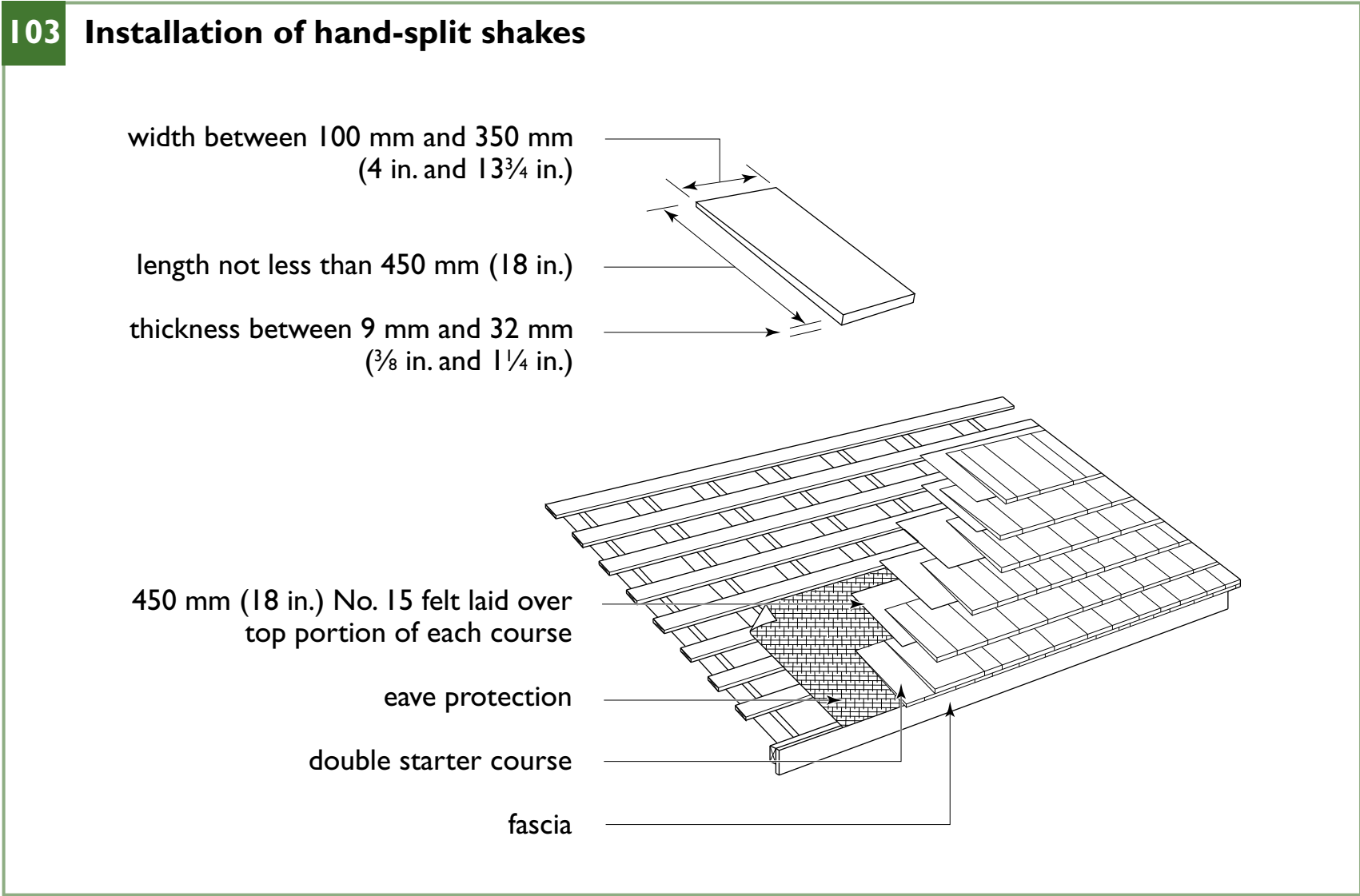
Cedar hand-split shakes must not be less than 450 mm (18 in.) long and 100 mm (4 in.) wide, and no wider than 350 mm (13¾ in.). The butt thickness should be between 9 and 32 mm (¾ in. and 1¼ in.) (Figure 103).

Shakes may be applied over spaced or closed roof sheathing. When spaced sheathing is used (Figure 98), apply 19 × 89 mm (1 × 4 in. nominal) or wider strips on centre equal to the weather exposure at which the shakes are to be laid, but never more than 250 mm (10 in.). Closed roof sheathing is recommended for areas where wind-driven snow conditions prevail.

An exposure of 190 mm (7½ in.) is recommended for 450 mm (18 in.) long shakes, and 250 mm (10 in.) for 600 mm (24 in.) long shakes. The minimum recommended roof pitch for hand-split shakes is 1:3.

A 900 mm (36 in.) strip of No. 15 roofing felt should be laid over the sheathing boards at the eave line. The beginning or starter course of shakes should be doubled; for extra texture, it can be tripled. The bottom course or courses can be 380 mm (15 in.) or 450 mm (18 in.) shakes.

After each course of shakes is applied, a 450 mm (18 in.) wide strip of No. 15 roofing felt should be laid over the top portion of the shakes, extending onto the sheathing. The bottom edge of the felt should be positioned above the butt at a distance equal to twice the weather exposure. For example, 600 mm (24 in.)



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Roof Sheathing and Coverings

shakes laid with 250 mm (10 in.) exposure would need to have felt applied 500 mm (20 in.) above the shake butts. Thus, the felt will cover the top 100 mm (4 in.) of the shakes and extend about 350 mm (15 in.) onto the sheathing (**Figure 103**).

Space shakes from 6 to 9 mm ($\frac{1}{4}$ to $\frac{3}{8}$ in.) apart. Offset side joints no less than 40 mm ($1\frac{5}{16}$ in.) over the adjacent courses. When straight split shakes are used, the “fro-e-end” (the end from which the shakes have been split, and which is smoother) should be laid uppermost (towards the ridge).

Finish at Ridge and Hips

The most common type of finish for asphalt shingles is shown in **Figure 104**. Asphalt shingle squares (one third of a strip) are used over the

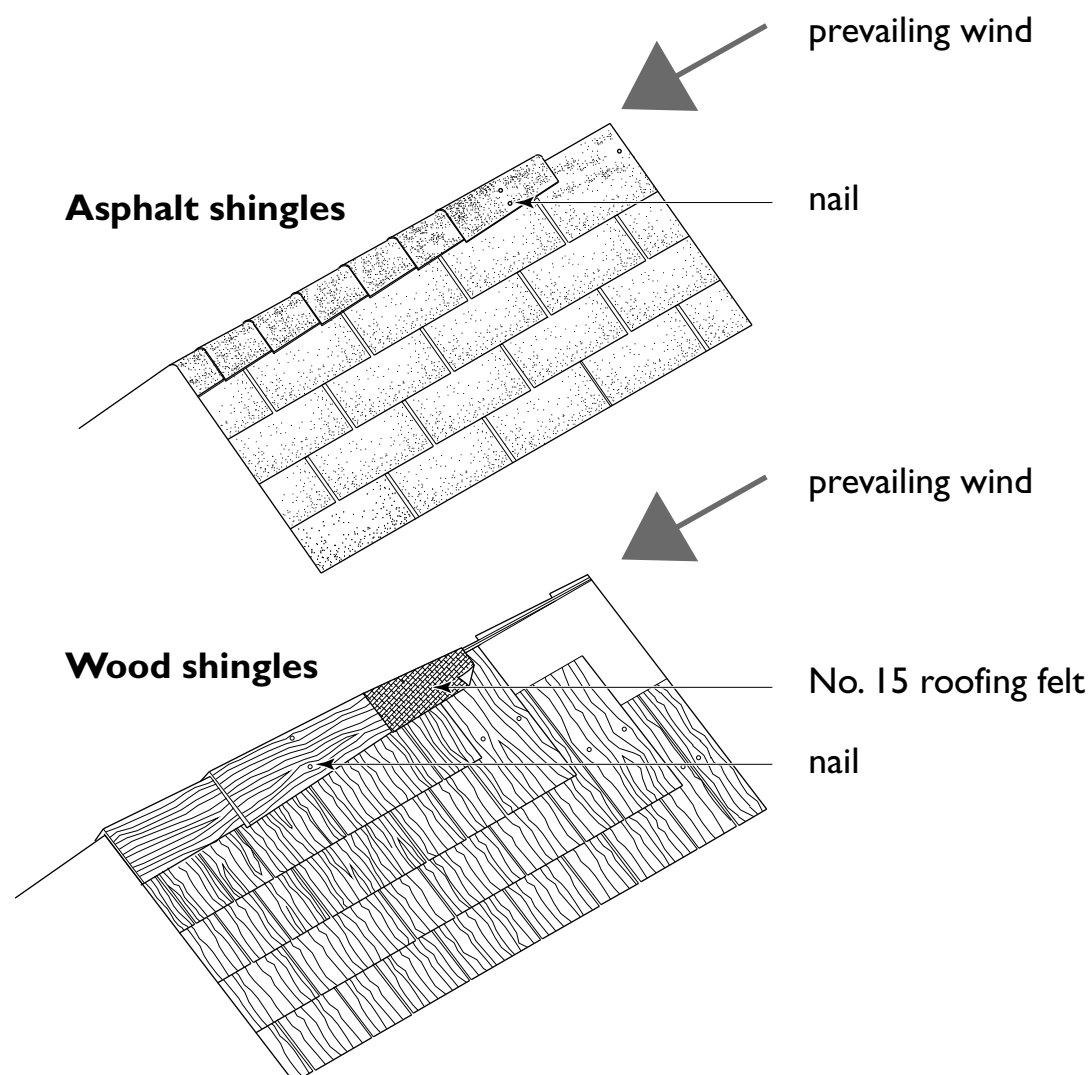
ridge or hip and blind-nailed. Each shingle is lapped to provide the same coverage as the roofing shingles. Orient the ridge cap to provide maximum protection from the prevailing wind.

For wood shingles, 150 mm (6 in.) wide shingles are alternatively lapped and blind-nailed (**Figure 104**). Flashing is sometimes used under a wood-shingle ridge.

Built-up Roofs

Built-up roof coverings should be installed by specialized roofing installers. A built-up roof may have three or more layers of roofing felt. Each layer, including the final layer, is sealed with tar or asphalt. The surface is then covered with gravel (embedded in the asphalt or tar) or with a cap sheet. This covering provides ballast against wind uplift and protection from the sun.

104 Finish at ridge and hips



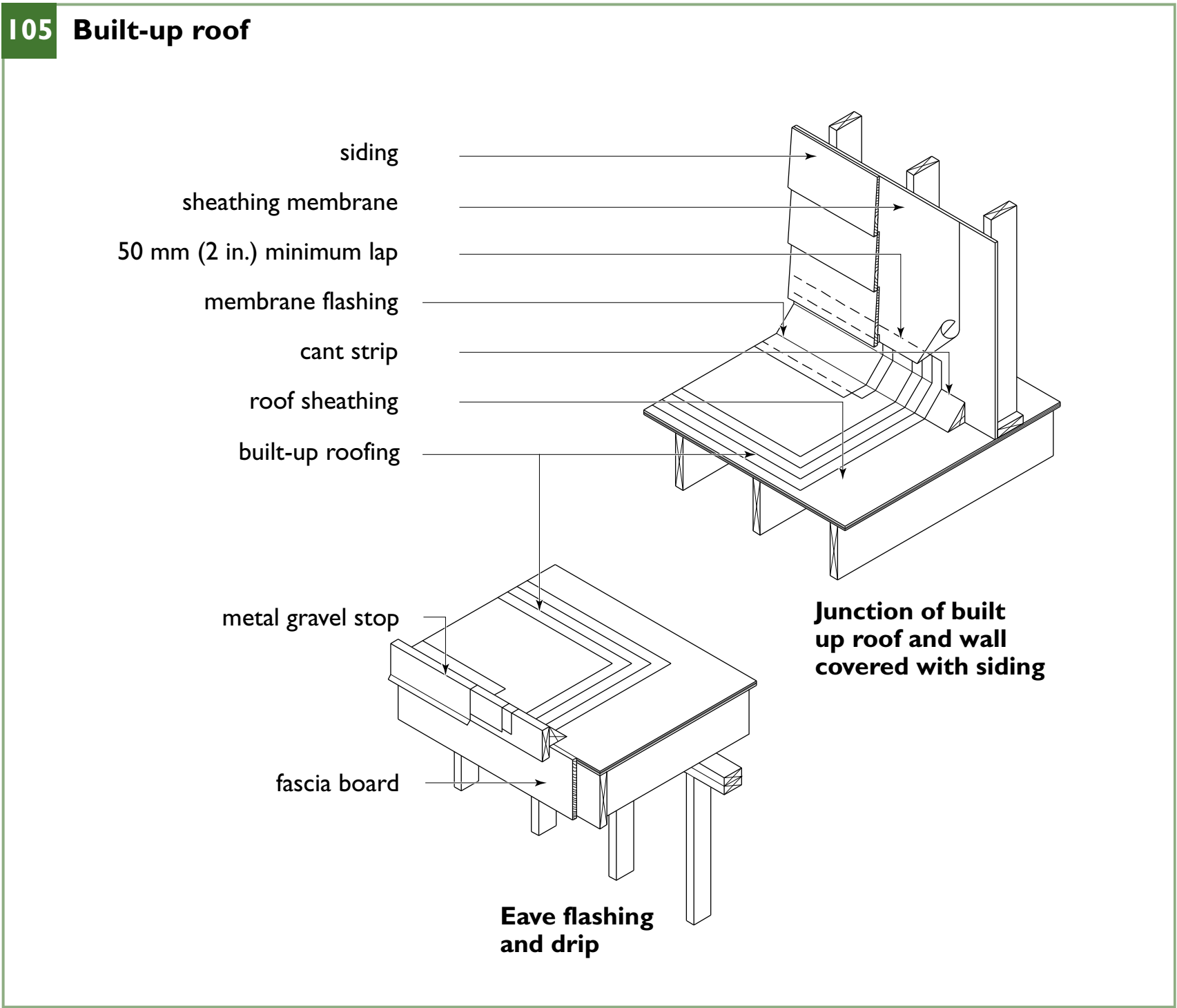
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Coal tar products and asphalt products are not compatible and must not be used together.

The eave line of projecting roofs is trimmed with metal edging or flashing to direct water away from the fascia board. A gravel stop or cant strip is used in conjunction with the flashing at the eaves when the roof is covered with gravel (Figure 105). Where built-up roofing abuts a wall (except a masonry-clad wall), the roofing is mopped to the cant strip and turned up the wall at least 150 mm (6 in.). The wall sheathing paper and siding is then lapped over the roof membrane).

Where a built-up roof intersects a masonry-clad wall, the roof membrane is turned up the face of the masonry and a flashing is installed over the membrane. A counter-flashing should be embedded into the mortar joints at least 25 mm (1 in.), extend down the wall about 150 mm (6 in.) and lap over the membrane flashing at least 100 mm (4 in.).

Single-ply membranes can also be used for low-slope roofs including the small roofs typical of wood-frame construction.



Sheet Metal Roofing

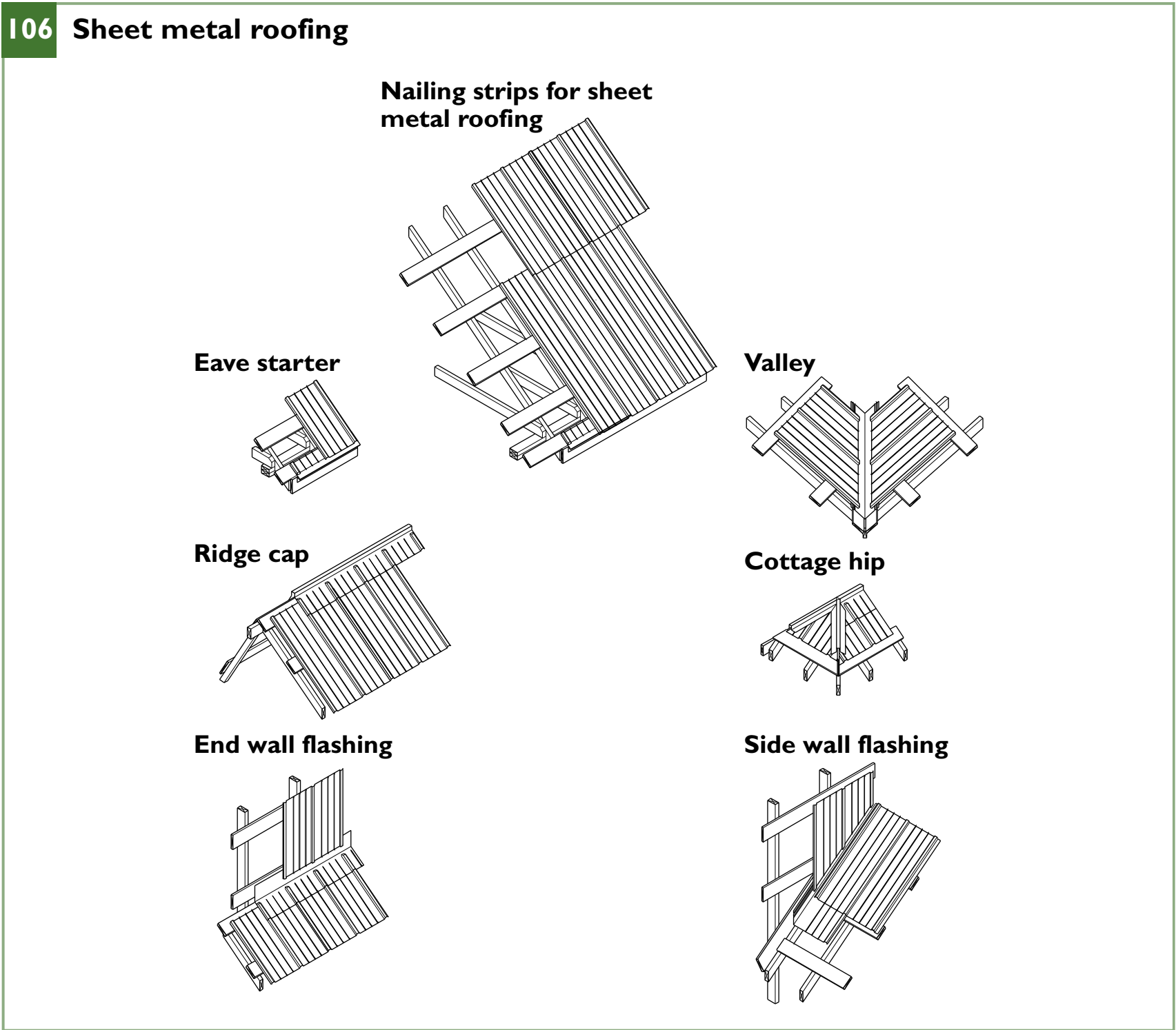
Sheet metal roofing is manufactured in widths from 762 to 914 mm (30 to 36 in.) and in any lengths specified by the builder. Various accessories are needed for hips and valleys, eave starters and edges (Figure 106).

The minimum slope for sheet metal roofing is 1:4, unless the manufacturer provides written confirmation that a lower slope has been tested and proven. If sheet metal roofing is applied over spaced supports, the sheets need to be designed to support the snow load, and there must be a support under each end joint (Figure 106).

The choice of metal thickness, whether steel or aluminum, depends on the local snow load and the profile of the metal roofing, but should not be less than 0.33 mm (0.013 in.) for galvanized steel, 0.46 mm (0.018 in.) for copper or zinc and 0.48 mm (0.019 in.) for aluminum. The required thicknesses for specific snow loads are provided by the manufacturer.

Concrete and Clay Tile Roofing

Concrete or clay tile is considerably heavier than other roofing systems and the roof structure must be designed by an engineer to withstand the additional load.



SUSTAINABLE HOUSING INSIGHT

Healthy Indoor Environment

- Design and construct the roofing system to deflect all water away from the roof and walls, and to prevent water from infiltrating the living spaces. Moisture in ceilings, walls and floors can promote mold growth, which could lead to health problems among occupants.
- Project water-shedding beyond materials below, and provide drip edges to deflect water outwards and away from cladding materials, windows, doors and penetrations.

Energy Efficiency

- Consider designing and orienting the roof so that it may accommodate solar thermal and/or solar photovoltaic renewable energy production in future. Choose a roof finish with a service life equal to or greater than that of the renewable systems.
- Consider increasing roof sheathing thickness or reducing the spacing between roof framing members to accommodate future solar thermal and photovoltaic panels.
- Building integrated solar photovoltaics such as solar roof tiles can produce renewable energy without compromising the appearance of the roof.

Resource Efficiency and Affordability

- Select the type of roof sheathing that best suits local requirements for durability, availability, place of manufacture and transportation cost.
- Select a roof covering that has a good appearance, is durable and provides the best balance between initial cost and service life.

- Reduce roof waste. Estimate quantities carefully. As much as possible, make the dimensions of roof surfaces a multiple of 400 or 600 mm (16 or 24 in.) to reduce waste from panel roof sheathing.
- Metal roofing can be re-used or recycled after the end of the building's service life.
- Asphalt-fibreglass, metal, and concrete roof coverings are made from non-renewable materials. Wood products from a properly managed forest are renewable.
- The service life of asphalt-fibreglass shingles depends on their quality, roof slope and their exposure to sun and other weather conditions. Metal and concrete roof coverings can last significantly longer. Wood shingles can also last longer but this depends on many variables (drainage and drying, grain direction, susceptibility to splitting and curling, etc.).
- The service life of roof coverings can be maximized by providing conditions that allow them to dry between wettings. This means keeping roofs clear of leaves and branches and trimming overhanging branches to permit drying and prevent the growth of moss.
- Concrete roof coverings are very durable but heavy, and additional roof framing may be needed to provide adequate structural support. In areas with a high earthquake risk, additional structural reinforcement must be provided to support heavy roofing materials.